

宁波大学
NINGBO UNIVERSITY

云计算技术实验报告

[鲲鹏云上应用高可用部署]

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实验项目名称	鲲鹏云上应用高可用部署

一、实验目的与要求：

关于实验：

本实验将介绍如何在鲲鹏平台上部署企业自动化办公（Office Automation，简称 OA）系统，如何将本地自建的 MySQL 数据库迁移到云数据库 GaussDB，以及如何在鲲鹏平台上部署应用的高可靠。

实验目的：

- 掌握华为云鲲鹏云服务的操作方法；
- 掌握自建数据库迁移到云数据库 GaussDB 的原理、流程和操作方法；
- 完成鲲鹏云平台应用的高可靠部署。

鲲鹏平台部署 OA 系统

- 掌握应用在华为云鲲鹏云服务器上的部署和配置流程。
- 掌握华为云数据库的部署和配置流程。

鲲鹏平台数据库迁移与部署

- 理解 Java 应用连接数据库的基本原理。
- 掌握数据库数据迁移的基本原理和操作。
- 掌握部署 GaussDB(for MySQL)的方法。

鲲鹏平台应用高可用部署

- 了解和使用华为云弹性负载均衡 ELB 和弹性伸缩 AS 服务。
- 掌握在鲲鹏平台部署高可用架构的方法

二、实验环境：

实验设备详细信息

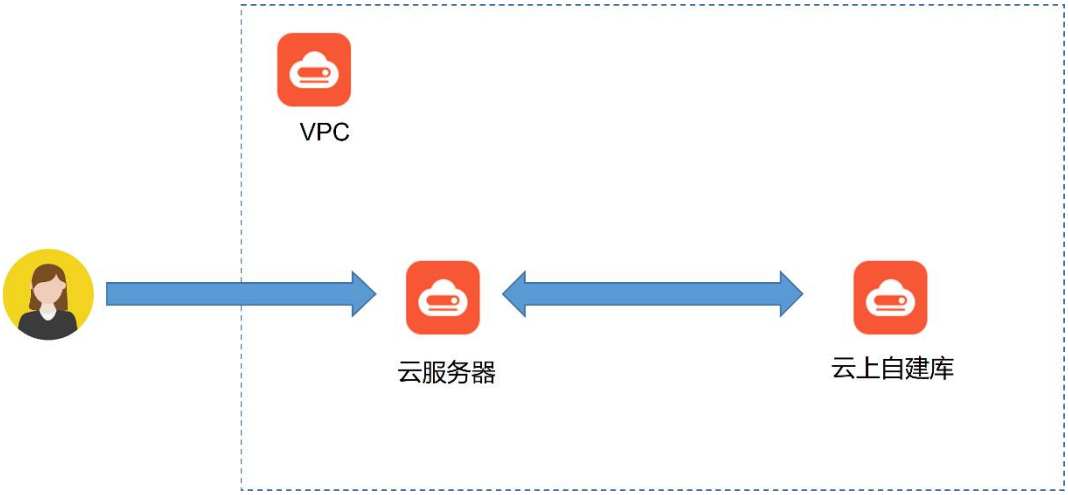
设备名称	设备型号	软件版本
鲲鹏平台MySQL服务器 ecs-mysql	鲲鹏计算，通用计算增强型， kc1.large.2 | 2vCPUs | 4GB	openEuler 20.03 64bit
鲲鹏平台OA系统服务器 ecs-oa	鲲鹏计算，通用计算增强型， kc1.2xlarge.2 | 8vCPUs | 16GB	openEuler 20.03 64bit with ARM
云数据库GaussDB(for MySQL)gauss-oa	GaussDB(for MySQL)，通用增强型，16vCPUs | 64 GB	兼容MySQL 8.0

实验工具详细信息

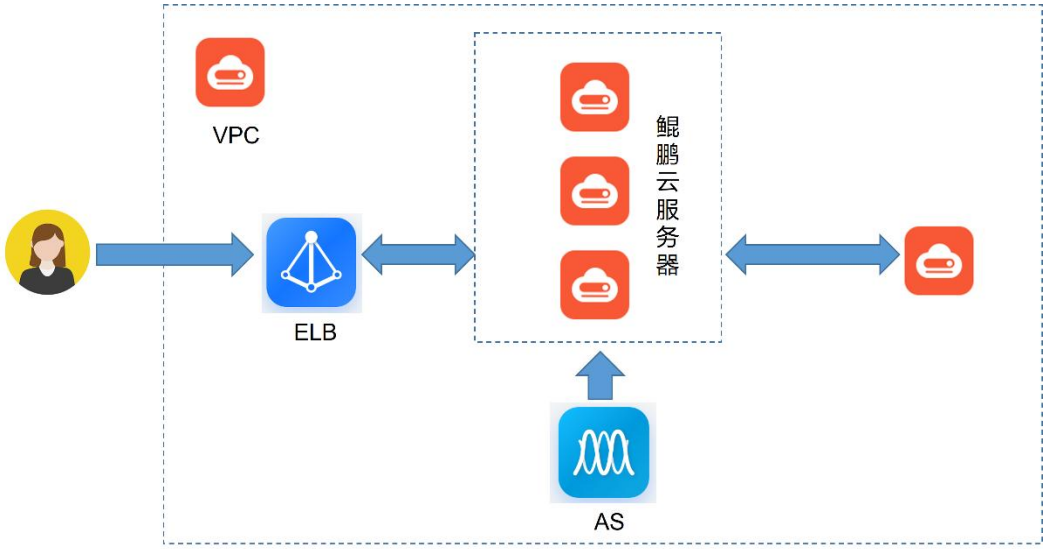
名称	下载链接	用途
PuTTY	win64 位：https://the.earth.li/~sgtatham/putty/latest/w64/putty-64bit-0.74-installer.msi	远程登录工具
WinSCP	https://winscp.net/eng/download.php	远程传输工具
Git	https://git-scm.com/downloads	分布式版本控制系统
Maven	https://mirrors.huaweicloud.com/apache/maven/maven-3/3.6.3/binaries/apache-maven-3.6.3-bin.tar.gz	软件项目管理工具
oasys_mysql	https://gitee.com/github-5407963/oasys_mysql	OA系统项目源代码
oasys_mysql.sql	https://obs-oa-zyl.obs.cn-north-4.myhuaweicloud.com/oasys_mysql.sql	OA系统MySQL数据库脚本

三、实验方法（原理、流程图）

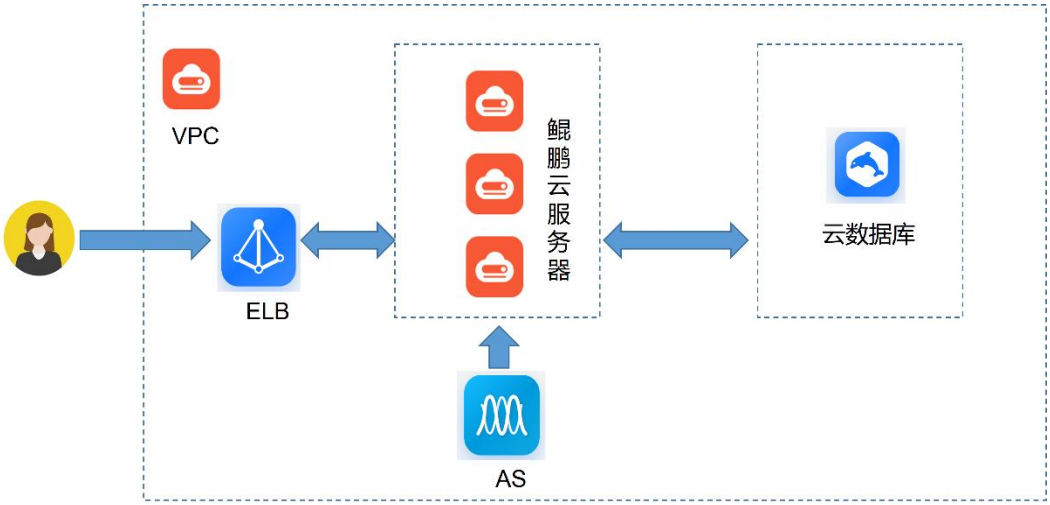
实验一：鲲鹏平台部署 OA 系统架构图



实验二：鲲鹏应用迁移系统架构图



实验三：鲲鹏数据库迁移系统架构图



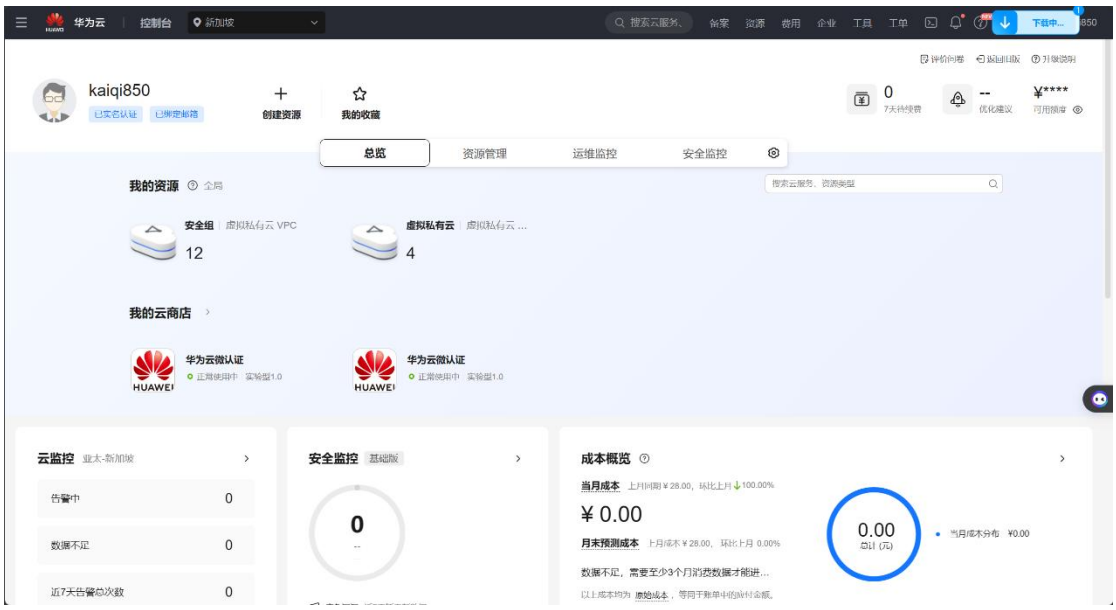
四、实验过程、步骤及内容

目录

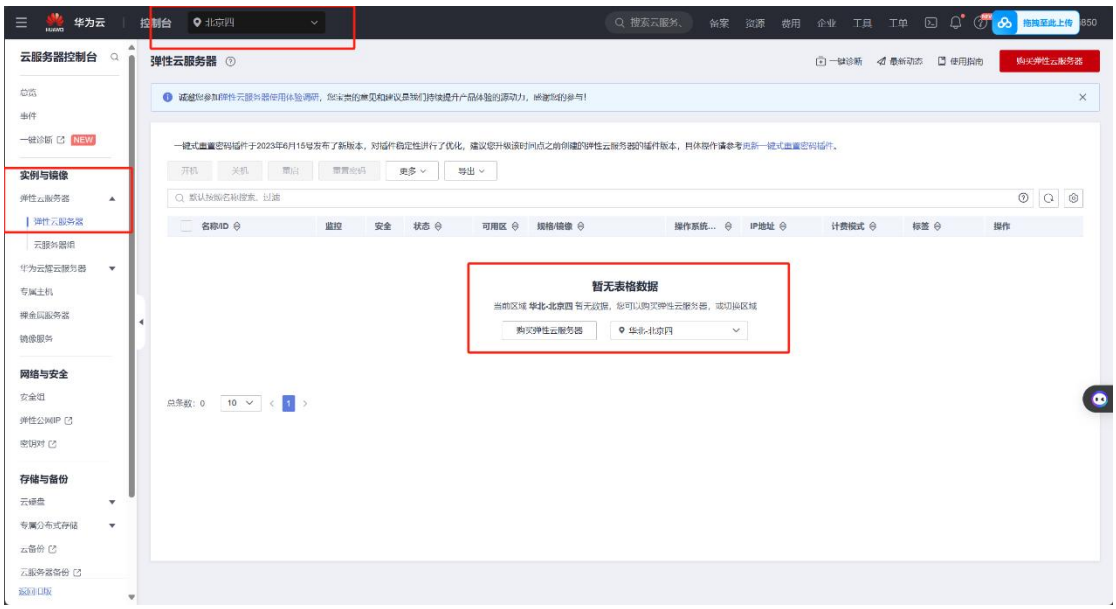
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一：弹性云服务器 ECS 部署

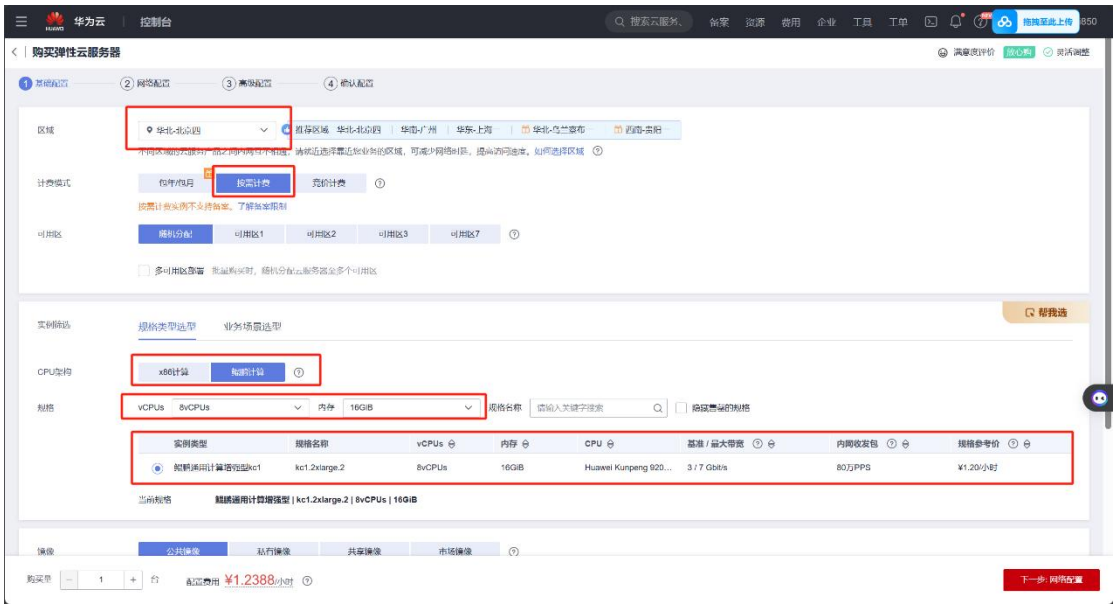
在华为云首页， 单击右上角的“控制台”，进入控制台操作页面



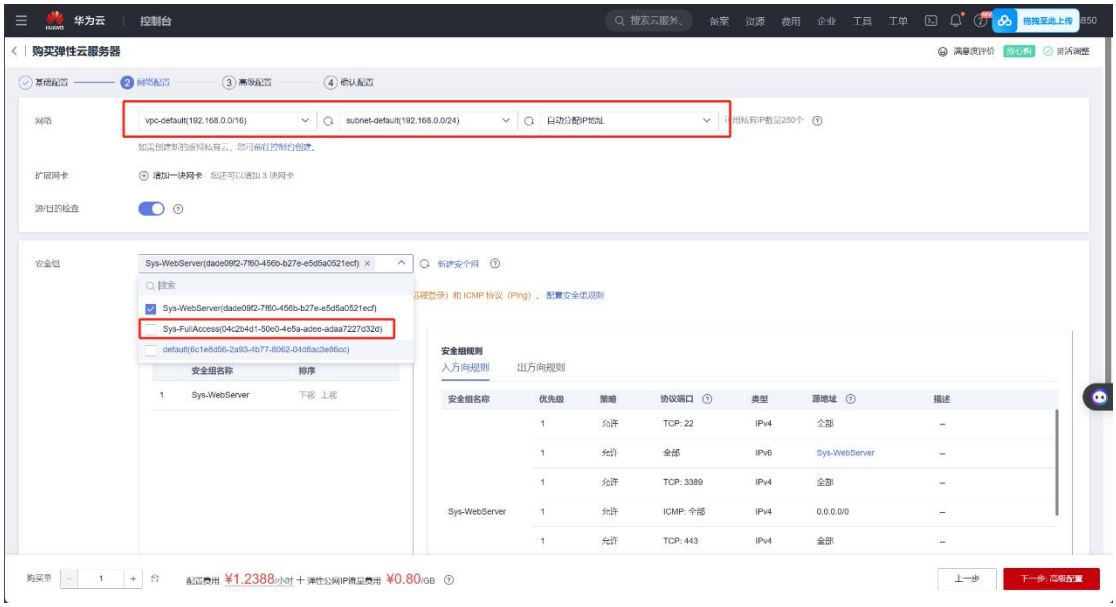
在页面左上角，选择区域“北京四” ，单击“服务列表 ”，选择“计算 -> 弹性云服务器 ECS” ，进入弹性云服务器列表页面。点击购买弹性云服务器 ECS。



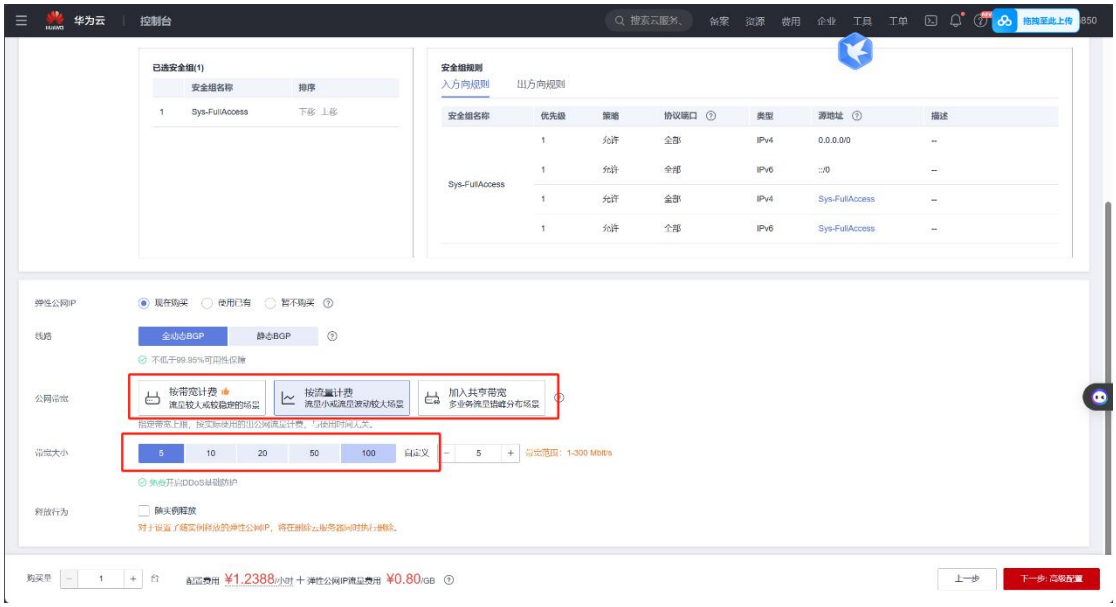
进入弹性云服务器的基础配置页面， 选择基础配置（按需计费，鲲鹏计算,鲲鹏通用计算增强型，kc1.2xlarge.2 8vCPUs | 16GB ,openEuler 20.03 64bit)， 单击“下一步：网 络配置 ”。



进入弹性云服务器的网络配置页面， 其中网络选择“vpc-default -> subnet-default -> 自动分配地址”，网络配置如下图所示：



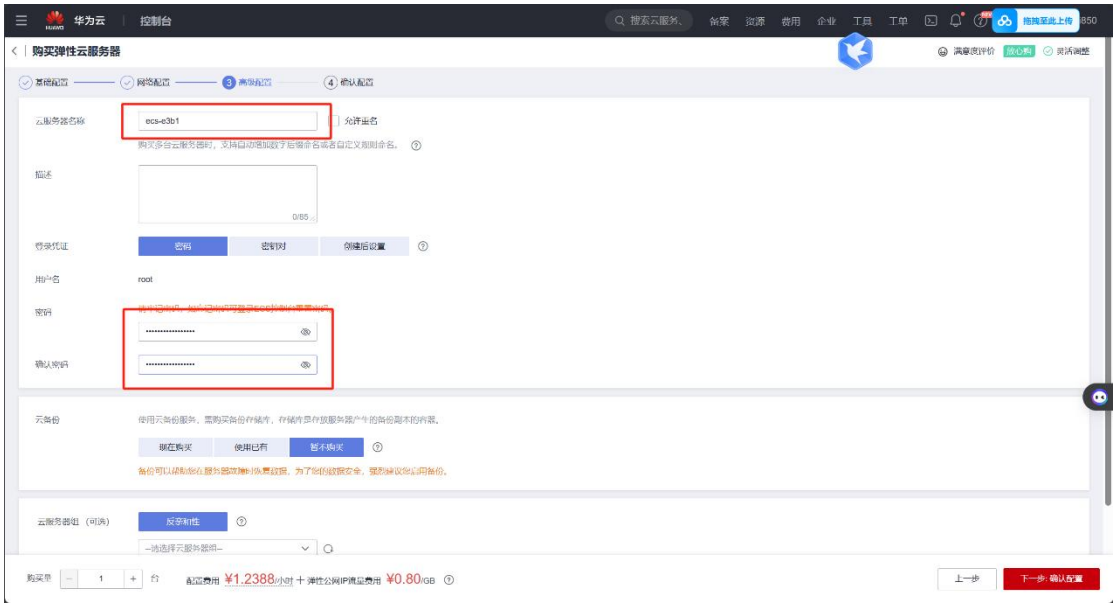
点击安全组刷新按钮，选择“sg-FullAccess”，公网带宽选择“按流量计费”和“5”Mbit/s 带宽大小，单击“下一步：高级配置 ”。



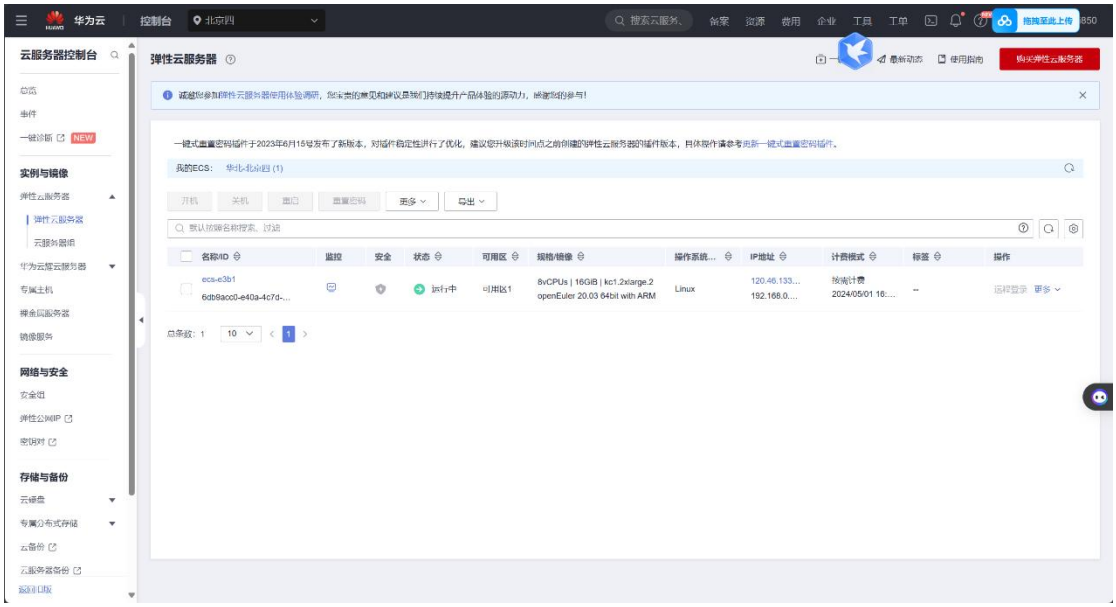
进入弹性云服务器的高级配置页面，高级配置如下图所示：

设置账号密码为：Jklzxcbnm1597538@

单击“下一步：确认配置 ”。

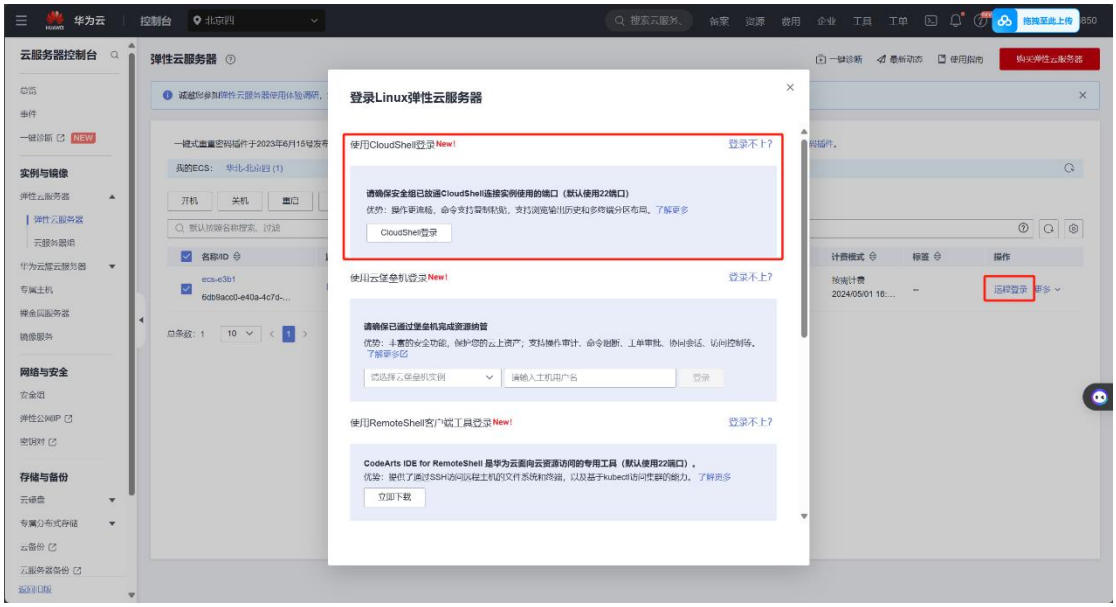


返回弹性云服务器列表页面，查看刚刚购买的弹性云服务，等待其状态变为“运行中”， 记录其弹性 IP 地址。



二：安装 Cmake

在本地 PC 中打开 PuTTY 或其他 SSH 登录工具， 也可以使用云服务器右侧“远程登录 ”中的“CloudShell 登录 ”。



检查 CMake 是否安装， 返回内容如下所示，表示未安装 cmake ，需要安装 cmake。

```
[root@ecs-e3b1 ~]# rpm -qa | grep cmake
[root@ecs-e3b1 ~]#
```

执行以下命令，安装所需依赖包。

```
[root@ecs-e3b1 ~]# dnf -y install gcc gcc-c++ automake zlib zlib-devel bzip2 bzip2-devel
Last metadata expiration check: 0:00:18 ago on Wed 01 May 2024 05:09:02 PM CST.
Package gcc-7.3.0-20210605.41.oe1.aarch64 is already installed.
Package zlib-1.2.11-17.oe1.aarch64 is already installed.
Package bzip2-1.0.8-3.oe1.aarch64 is already installed.
Dependencies resolved.
================================================================================================================================================================
 Package                                Architecture                    Version                                 Repository                            Size
================================================================================================================================================================
Installing:
 bzip2-devel                            aarch64                         1.0.8-3.oe1                             OS                                    38 k
 zlib-devel                             aarch64                         1.2.11-17.oe1                           OS                                    97 k
 automake                               noarch                          1.16.2-1.oe1                            update                               452 k
 gcc-c++                                aarch64                         7.3.0-20210605.41.oe1                   update                               7.3 M
Installing dependencies:
 autoconf                               noarch                          2.69-30.oe1                             OS                                   610 k
 m4                                     aarch64                         1.4.18-13.oe1                           OS                                    88 k
 emacs-filesystem                       noarch                          1:26.1-13.oe1                           update                               8.0 k
 libstdc++-devel                        aarch64                         7.3.0-20210605.41.oe1                   update                               1.1 M

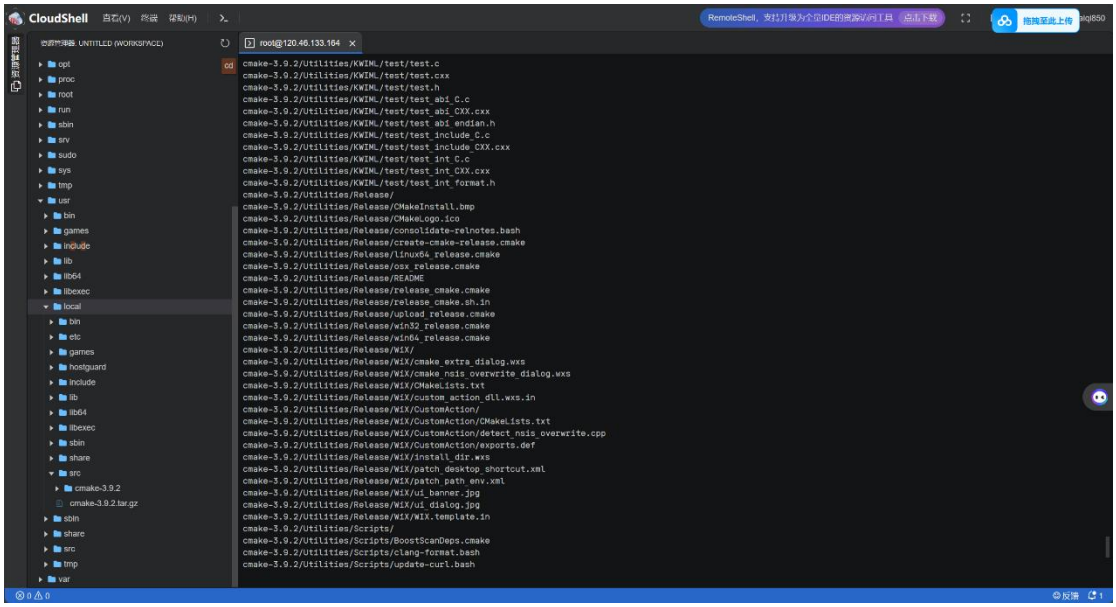
Transaction Summary
================================================================================================================================================================
Install  8 Packages

Total size: 9.6 M
Installed size: 34 M
Downloading Packages:
[SKIPPED] autoconf-2.69-30.oe1.noarch.rpm: Already downloaded
[SKIPPED] bzip2-devel-1.0.8-3.oe1.aarch64.rpm: Already downloaded
[SKIPPED] m4-1.4.18-13.oe1.aarch64.rpm: Already downloaded
[SKIPPED] zlib-devel-1.2.11-17.oe1.aarch64.rpm: Already downloaded
[SKIPPED] automake-1.16.2-1.oe1.noarch.rpm: Already downloaded
[SKIPPED] emacs-filesystem-26.1-13.oe1.noarch.rpm: Already downloaded
[SKIPPED] gcc-c++-7.3.0-20210605.41.oe1.aarch64.rpm: Already downloaded
[SKIPPED] libstdc++-devel-7.3.0-20210605.41.oe1.aarch64.rpm: Already downloaded
Package autoconf-2.69-30.oe1.noarch.rpm is not signed
Package bzip2-devel-1.0.8-3.oe1.aarch64.rpm is not signed
Package m4-1.4.18-13.oe1.aarch64.rpm is not signed
```

获取 CMake 源码，使用如下指令产生报错（地址过期）具体解决过程见[第五部分 Q1&&S1]

```
[root@ecs-e3b1 ~]# cd /usr/local/src
[root@ecs-e3b1 src]# wget https://zhuanyejianshe.obs.cn-north-
--2024-05-01 17:11:47--  https://zhuanyejianshe.obs.cn-north-/
Resolving zhuanyejianshe.obs.cn-north- (zhuanyejianshe.obs.cn-north-)... failed: Name or service not known.
wget: unable to resolve host address 'zhuanyejianshe.obs.cn-north-'
[root@ecs-e3b1 src]# 4.myhuaweicloud.com/yunjisuan-gongjuruanjian/cmake-3.9.2.tar.gz
-bash: 4.myhuaweicloud.com/yunjisuan-gongjuruanjian/cmake-3.9.2.tar.gz: No such file or directory
```

解压软件包:



进入 CMake 的安装目录。

```
[root@ecs-e3b1 src]# cd cmake-3.9.2
[root@ecs-e3b1 cmake-3.9.2]#
```

安装 CMake

```
[root@ecs-e3b1 cmake-3.9.2]# ./configure
-- ------------------------------------------
CMake 3.9.2, Copyright 2000-2017 Kitware, Inc. and Contributors
Found GNU toolchain
C compiler on this system is: gcc  -std=gnu11
C++ compiler on this system is: g++  -std=gnu++14
Makefile processor on this system is: gmake
g++ is GNU compiler
g++ has setenv
g++ has unsetenv
g++ does not have environ in stdlib.h
g++ has stl wstring
g++ has <ext/stdio_filebuf.h>
-- ------------------------------------------
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddCustomCommandCommand.cxx -o cmAddCustomCommandCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddCustomTargetCommand.cxx -o cmAddCustomTargetCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddDefinitionsCommand.cxx -o cmAddDefinitionsCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddDependenciesCommand.cxx -o cmAddDependenciesCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddExecutableCommand.cxx -o cmAddExecutableCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddLibraryCommand.cxx -o cmAddLibraryCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddSubDirectoryCommand.cxx -o cmAddSubDirectoryCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmAddTestCommand.cxx -o cmAddTestCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
/cmake-3.9.2/Utilities  -c /usr/local/src/cmake-3.9.2/Source/cmBreakCommand.cxx -o cmBreakCommand.o
g++  -std=gnu++14     -I/usr/local/src/cmake-3.9.2/Bootstrap.cmk   -I/usr/local/src/cmake-3.9.2/Source   -I/usr/local/src/cmake-3.9.2/Source/LexerParser   -I/usr/local/src
```

```
[root@ecs-e3b1 cmake-3.9.2]# make -j8 && make install
make[1]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
Scanning dependencies of target cmcompress
Scanning dependencies of target cmsys_c
Scanning dependencies of target cmlibrhash
Scanning dependencies of target kwiml_test
Scanning dependencies of target cmsys
Scanning dependencies of target cmexpat
Scanning dependencies of target cmzlib
make[2]: Leaving directory '/usr/local/src/cmake-3.9.2'
Scanning dependencies of target cmbzip2
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Leaving directory '/usr/local/src/cmake-3.9.2'
[  0%] Building C object Utilities/cmcompress/CMakeFiles/cmcompress.dir/cmcompress.c.o
make[2]: Leaving directory '/usr/local/src/cmake-3.9.2'
make[2]: Leaving directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Leaving directory '/usr/local/src/cmake-3.9.2'
make[2]: Leaving directory '/usr/local/src/cmake-3.9.2'
make[2]: Leaving directory '/usr/local/src/cmake-3.9.2'
[  0%] Building C object Utilities/KWIML/test/CMakeFiles/kwiml_test.dir/test.c.o
[  0%] Building C object Utilities/cmlibrhash/CMakeFiles/cmlibrhash.dir/librhash/algorithms.c.o
[  0%] Building C object Utilities/cmbzip2/CMakeFiles/cmbzip2.dir/blocksort.c.o
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
make[2]: Entering directory '/usr/local/src/cmake-3.9.2'
```

测试 CMake 是否安装完成， 返回内容如下所示，表示安装已经完成。

```
[root@ecs-e3b1 cmake-3.9.2]# cmake -version
cmake version 3.9.2

CMake suite maintained and supported by Kitware (kitware.com/cmake).
[root@ecs-e3b1 cmake-3.9.2]#
```

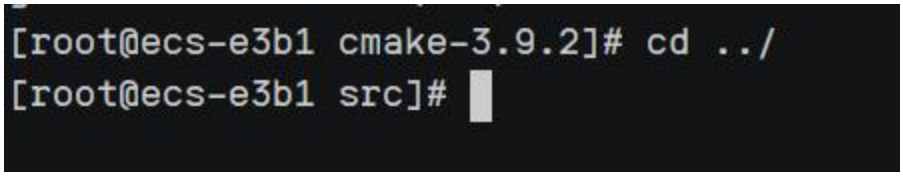
检查GCC 编译器版本

```
[root@ecs-e3b1 cmake-3.9.2]# gcc -v
Using built-in specs.
COLLECT_GCC=gcc
COLLECT_LTO_WRAPPER=/usr/libexec/gcc/aarch64-linux-gnu/7.3.0/lto-wrapper
Target: aarch64-linux-gnu
Configured with: ../configure --prefix=/usr --mandir=/usr/share/man --infodir=/usr/share/info --enable-shared --enable-threads=posix --enable-checking=release --with-syste
m-zlib --enable-__cxa_atexit --disable-libunwind-exceptions --enable-gnu-unique-object --enable-linker-build-id --with-linker-hash-style=gnu --enable-languages=c,c++,objc,
obj-c++,fortran,lto --enable-plugin --enable-initfini-array --disable-libgcj --with-isl=/home/abuild/rpmbuild/BUILD/gcc-7.3.0/obj-aarch64-linux-gnu/isl-install --without-c
loog --enable-gnu-indirect-function --build=aarch64-linux-gnu --with-stage1-ldflags=' -Wl,-z,relro,-z,now' --with-boot-ldflags=' -Wl,-z,relro,-z,now' --with-multilib-list=
lp64
Thread model: posix
gcc version 7.3.0 (GCC)
```

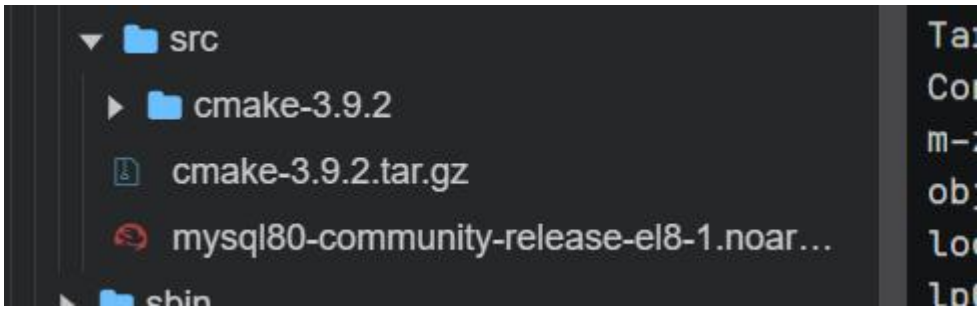
本实验 GCC 版本为 7.3.0 符合要求， 无需升级。

三：安装 MySQL

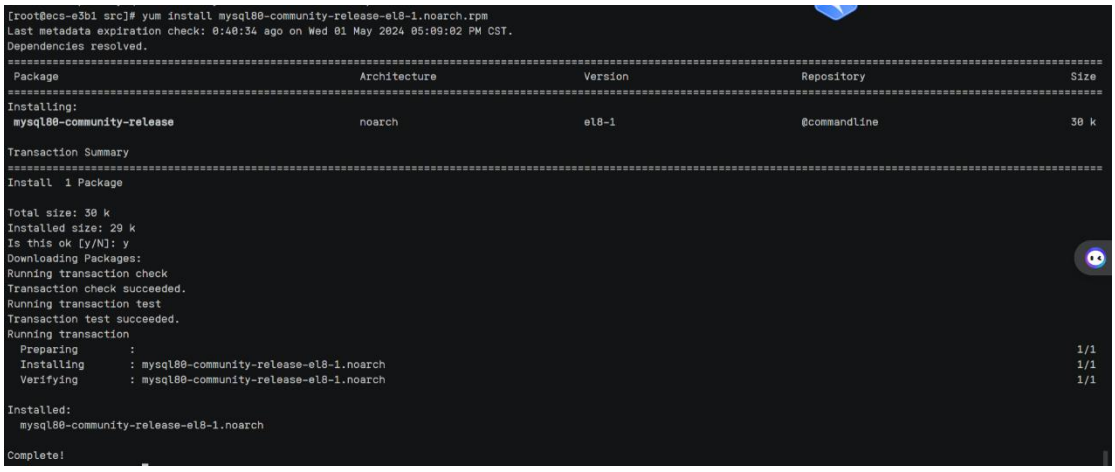
先退回到 src 文件夹



上传 SQL 文件



使用 yum install mysql80-community-release-el8-1.noarch.rpm 命令安装 SQL



下载服务：

```
[root@ecs-e3b1 src]# yum install mysql-community-server --nogpgcheck
Last metadata expiration check: 0:01:23 ago on Wed 01 May 2024 05:50:52 PM CST.
Dependencies resolved.
================================================================================
 Package                    Architecture  Version                 Repository      Size
================================================================================
Installing:
 mysql-community-server     aarch64       8.0.37-1.el8            mysql80-community  63 M
Installing dependencies:
 mysql-community-client     aarch64       8.0.37-1.el8            mysql80-community  15 M
 mysql-community-client-plugins aarch64   8.0.37-1.el8            mysql80-community 3.5 M
 mysql-community-common     aarch64       8.0.37-1.el8            mysql80-community 668 k
 mysql-community-icu-data-files aarch64   8.0.37-1.el8            mysql80-community 2.2 M
 mysql-community-libs       aarch64       8.0.37-1.el8            mysql80-community 1.4 M
 libaio                     aarch64       0.3.112-1.oe1           update            22 k

Transaction Summary
================================================================================
Install  7 Packages

Total download size: 86 M
Installed size: 411 M
Is this ok [y/N]: y
Downloading Packages:
(1/7): mysql-community-common-8.0.37-1.el8.aarch64.rpm          333 kB/s | 668 kB     00:02
(2/7): mysql-community-icu-data-files-8.0.37-1.el8.aarch64.rpm  1.5 MB/s | 2.2 MB     00:01
(3/7): mysql-community-client-plugins-8.0.37-1.el8.aarch64.rpm  714 kB/s | 3.5 MB     00:04
(4/7): mysql-community-client-8.0.37-1.el8.aarch64.rpm          916 kB/s |  15 MB     00:17
(5/7): libaio-0.3.112-1.oe1.aarch64.rpm                         505 kB/s |  22 kB     00:00
(6/7): mysql-community-libs-8.0.37-1.el8.aarch64.rpm             62 kB/s | 1.4 MB     00:23
(7/7): mysql-community-server-8.0.37-1.el8.aarch64.rpm  61% [=================================== ] 332 kB/s |  53 MB     01:41 ETA
```

启动服务

```
[root@ecs-e3b1 src]# /bin/systemctl start mysqld.service
[root@ecs-e3b1 src]#
```

检查服务状态：

```
[root@ecs-e3b1 src]# service mysqld status
Redirecting to /bin/systemctl status mysqld.service
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; vendor preset: disabled)
   Active: active (running) since Wed 2024-05-01 17:55:57 CST; 19s ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
  Process: 19448 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, status=0/SUCCESS)
 Main PID: 19518 (mysqld)
   Status: "Server is operational"
    Tasks: 38
   Memory: 473.5M
   CGroup: /system.slice/mysqld.service
           └─19518 /usr/sbin/mysqld

May 01 17:55:51 ecs-e3b1 systemd[1]: Starting MySQL Server...
May 01 17:55:57 ecs-e3b1 systemd[1]: Started MySQL Server.
```

获得初始化密码：ebb1lw,OU37g

```
[root@ecs-e3b1 src]# grep 'temporary password' /var/log/mysqld.log
2024-05-01T09:55:53.776564Z 6 [Note] [MY-010454] [Server] A temporary password is generated for root@localhost: ebb1lw,OU37g
```

使用命令 mysql -uroot -p 进入，输入初始密码

```
[root@ecs-e3b1 src]# mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.37

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

更新密码：

```
mysql> ALTER user 'root'@'localhost' IDENTIFIED BY'Jklzxcbnm1597538@';
Query OK, 0 rows affected (0.00 sec)
```

使用 flush privileges；刷新权限

使用 use mysql;

```
mysql> use mysql;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
```

使用 update user set host='%' where user ='root';

```
mysql> update user set host='%' where user ='root';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

使用 FLUSH PRIVILEGES;

```
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)
```

使用 GRANT ALL PRIVILEGES ON *.* TO 'root'@'%'WITH GRANT OPTION;

```
mysql> GRANT ALL PRIVILEGES ON *.* TO 'root'@'%'WITH GRANT OPTION;
Query OK, 0 rows affected (0.01 sec)
```

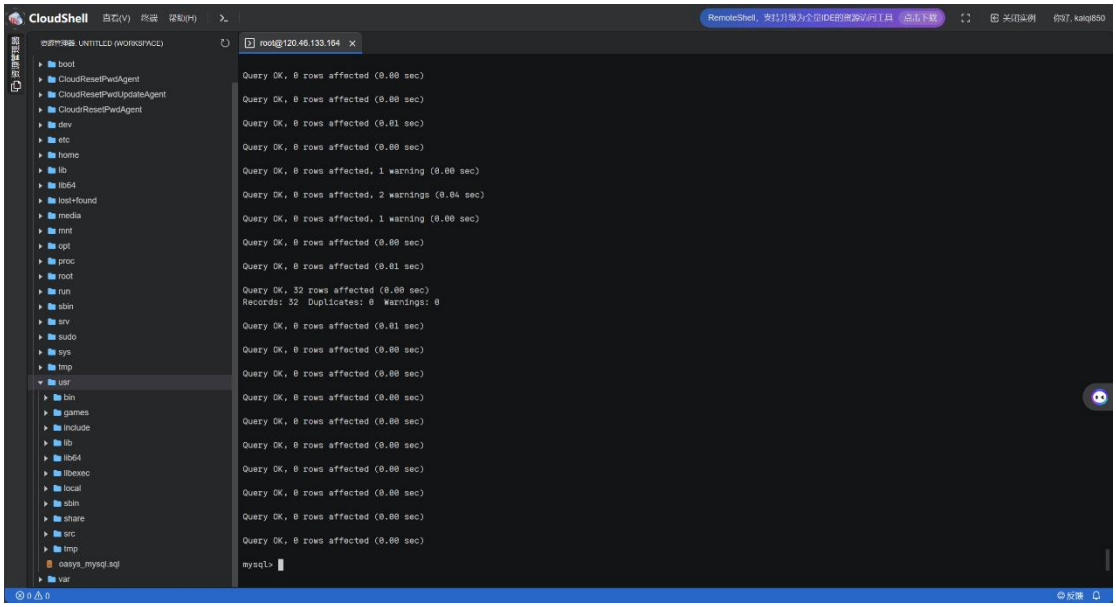
四：导入数据

下载原始sql 文件到/home 目录。

创建名为“oasys”的数据库。

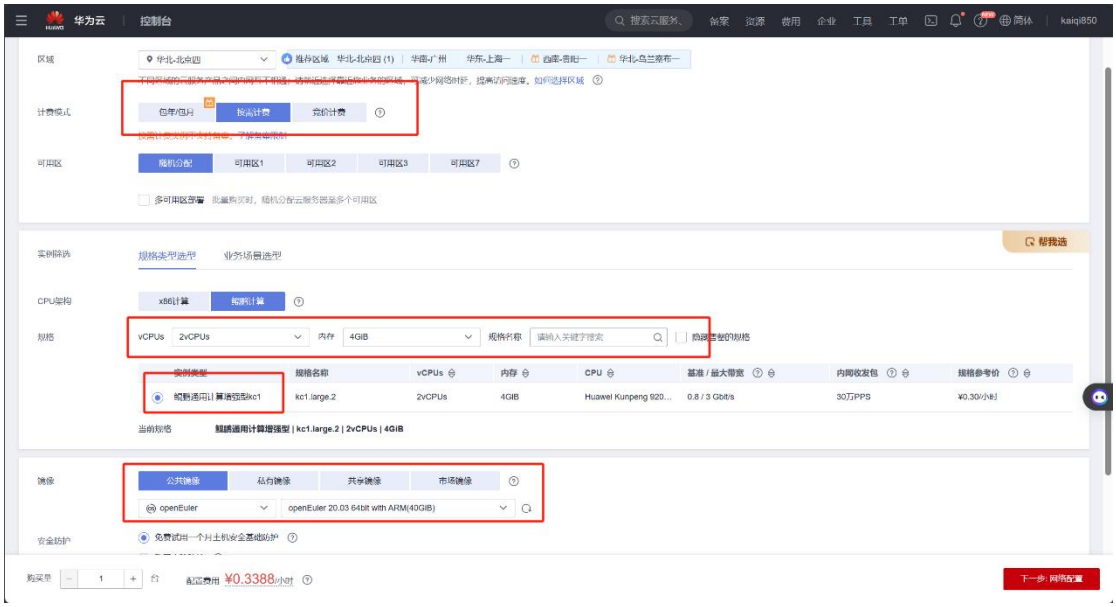
```
mysql>  create database oasys;
Query OK, 1 row affected (0.01 sec)
```

使用“oasys”数据库， 执行原始 sql 文件。



五：部署第二个云服务器

进入弹性云服务器的基础配置页面， 选择基础配置（按需计费，鲲鹏计算，鲲鹏通用计算增强型， kc1.large.2 2vCPUs | 4GB ，openEuler 20.03 64bit)， 单击“下一步：网络配 置 ”。

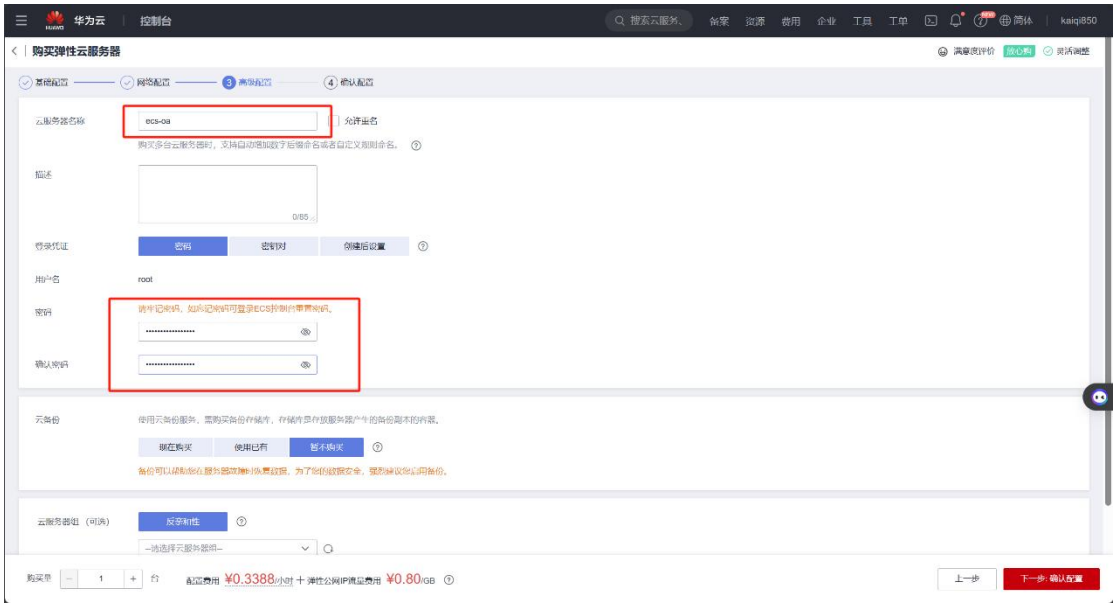


进入弹性云服务器的网络配置页面， 其中网络选择“vpc-default ->subnet-default->自动分配 ID 地址”,安全组选“sg-FullAccess”，公网带宽选择“按流量计费”和 “5”Mbit/s 带宽大小，单击“下一步：高级配置 ”。



进入弹性云服务器的高级配置页面，高级配置如下图所示：

密码为：Jklzxcbnm1597538@



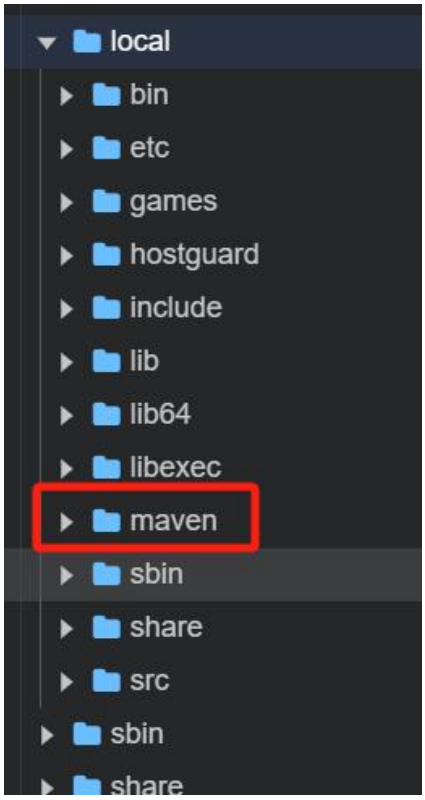
进入远程控制，输入 yum install java-1.8.0-openjdk* -y

－nogpgcheck　安装 JDK8

```
[root@ecs-oa ~]# yum install java-1.8.0-openjdk* -y --nogpgcheck
OS                                                                                                  24 MB/s | 3.2 MB     00:00
everything                                                                                          53 MB/s | 9.0 MB     00:00
EPOL                                                                                               6.6 MB/s | 891 kB     00:00
debuginfo                                                                                           23 MB/s | 2.7 MB     00:00
source                                                                                             5.6 MB/s | 816 kB     00:00
update                                                                                             1.5 MB/s |  16 MB     00:10
Dependencies resolved.
================================================================================================================================================================
 Package                                      Architecture                Version                                  Repository                             Size
================================================================================================================================================================
Installing:
 java-1.8.0-openjdk                           aarch64                     1:1.8.0.262.b10-0.oe1                    update                                356 k
 java-1.8.0-openjdk-debuginfo                 aarch64                     1:1.8.0.262.b10-0.oe1                    update                                108 M
 java-1.8.0-openjdk-debugsource               aarch64                     1:1.8.0.262.b10-0.oe1                    update                                7.9 M
 java-1.8.0-openjdk-demo                      aarch64                     1:1.8.0.262.b10-0.oe1                    update                                2.1 M
 java-1.8.0-openjdk-demo-slowdebug            aarch64                     1:1.8.0.262.b10-0.oe1                    update                                2.1 M
 java-1.8.0-openjdk-devel                     aarch64                     1:1.8.0.262.b10-0.oe1                    update                                9.7 M
 java-1.8.0-openjdk-devel-slowdebug           aarch64                     1:1.8.0.262.b10-0.oe1                    update                                9.7 M
 java-1.8.0-openjdk-headless                  aarch64                     1:1.8.0.262.b10-0.oe1                    update                                33 M
 java-1.8.0-openjdk-headless-slowdebug        aarch64                     1:1.8.0.262.b10-0.oe1                    update                                35
 java-1.8.0-openjdk-javadoc                   noarch                      1:1.8.0.262.b10-0.oe1                    update                                1
 java-1.8.0-openjdk-javadoc-zip               noarch                      1:1.8.0.262.b10-0.oe1                    update                                42
 java-1.8.0-openjdk-slowdebug                 aarch64                     1:1.8.0.262.b10-0.oe1                    update                                324 k
 java-1.8.0-openjdk-src                       aarch64                     1:1.8.0.262.b10-0.oe1                    update                                45 M
 java-1.8.0-openjdk-src-slowdebug             aarch64                     1:1.8.0.262.b10-0.oe1                    update                                45 M
Installing dependencies:
 copy-jdk-configs                             noarch                      3.7-3.oe1                                OS                                    23 k
 javapackages-filesystem                      noarch                      5.3.0-2.oe1                              OS                                    10 k
 libfontenc                                   aarch64                     1.1.4-2.oe1                              OS                                    18 k
 lua-posix                                    aarch64                     33.3.1-12.oe1                            OS                                    167 k
 xorg-x11-font-utils                          aarch64                     1:7.5-42.oe1                             OS                                    82 k
 xorg-x11-fonts-others                        noarch                      7.5-24.oe1                               OS                                    7.4 M
 alsa-lib                                     aarch64                     1.2.3-1.oe1                              update                                421 k
 lksctp-tools                                 aarch64                     1.0.18-1.oe1                             update                                62 k
 tzdata-java                                  noarch                      2020a-1.oe1                              update                                151 k
```

六：安装 Maven

执行以下命令，创建 Maven 安装目录。mkdir /usr/local/maven



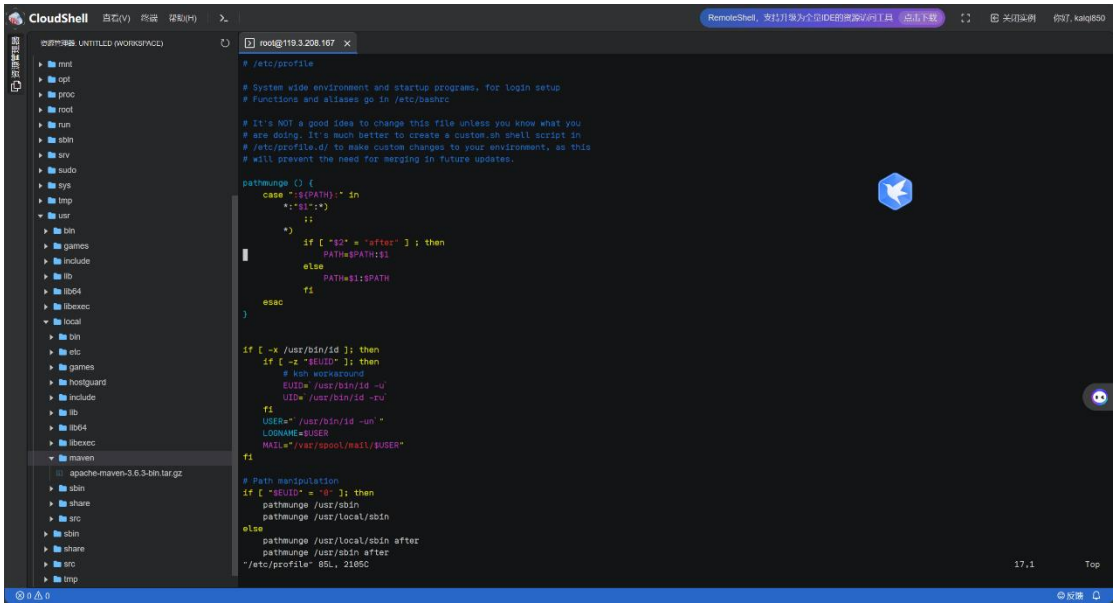
执行以下命令， 切换到 Maven 安装目录。

```
[root@ecs-oa ~]# cd /usr/local/maven
[root@ecs-oa maven]#
```

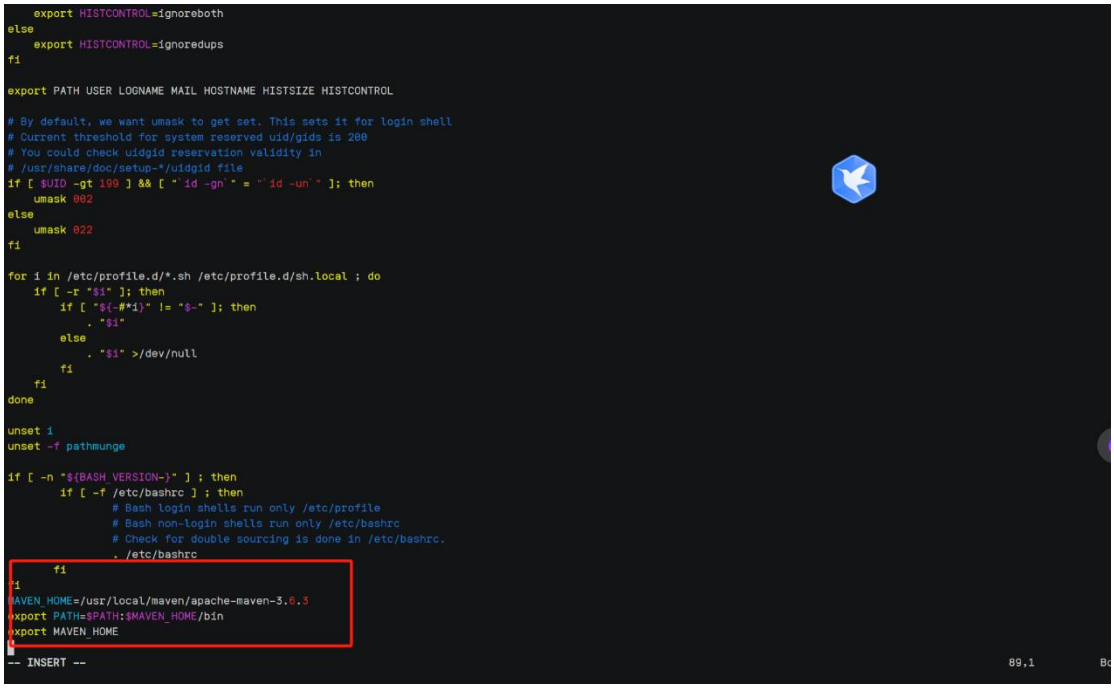
执行以下命令，获取 Maven 二进制包

```
[root@ecs-oa maven]# tar -xvzf apache-maven-3.6.3-bin.tar.gz
apache-maven-3.6.3/README.txt
apache-maven-3.6.3/LICENSE
apache-maven-3.6.3/NOTICE
apache-maven-3.6.3/lib/
apache-maven-3.6.3/lib/cdi-api.license
apache-maven-3.6.3/lib/commons-cli.license
apache-maven-3.6.3/lib/commons-io.license
apache-maven-3.6.3/lib/commons-lang3.license
apache-maven-3.6.3/lib/guava.license
apache-maven-3.6.3/lib/guice.license
apache-maven-3.6.3/lib/jansi.license
apache-maven-3.6.3/lib/javax.inject.license
apache-maven-3.6.3/lib/jcl-over-slf4j.license
apache-maven-3.6.3/lib/jsoup.license
apache-maven-3.6.3/lib/jsr250-api.license
apache-maven-3.6.3/lib/org.eclipse.sisu.inject.license
apache-maven-3.6.3/lib/org.eclipse.sisu.plexus.license
apache-maven-3.6.3/lib/plexus-cipher.license
apache-maven-3.6.3/lib/plexus-component-annotations.license
apache-maven-3.6.3/lib/plexus-interpolation.license
apache-maven-3.6.3/lib/plexus-sec-dispatcher.license
apache-maven-3.6.3/lib/plexus-utils.license
apache-maven-3.6.3/lib/slf4j-api.license
apache-maven-3.6.3/boot/
apache-maven-3.6.3/boot/plexus-classworlds.license
apache-maven-3.6.3/lib/jansi-native/
apache-maven-3.6.3/lib/jansi-native/freebsd32/
apache-maven-3.6.3/lib/jansi-native/freebsd64/
apache-maven-3.6.3/lib/jansi-native/linux32/
apache-maven-3.6.3/lib/jansi-native/linux64/
```

执行以下命令， 打开环境变量配置文件



按“i”进入编辑模式， 使用hjkl 键或方向键移动光标，在文件最后添加以下代码：



按“Esc”退出编辑模式，输入“:wq”并按回车，保存退出。

执行以下命令， 使新增配置生效。

```
[root@ecs-oa maven]# source /etc/profile


Welcome to 4.19.90-2110.8.0.0119.oe1.aarch64

System information as of time:  Wed May  1 21:24:18 CST 2024

System load:    0.00
Processes:      141
Memory used:    10.7%
Swap used:      0.0%
Usage On:       12%
IP address:     192.168.0.241
Users online:   1
```

执行以下命令，验证Maven 安装

```
[root@ecs-oa maven]# mvn -v
Apache Maven 3.6.3 (cecedd343002696d0abb50b32b541b8a6ba2883f)
Maven home: /usr/local/maven/apache-maven-3.6.3
Java version: 1.8.0_262, vendor: Boole, runtime: /usr/lib/jvm/java-1.8.0-openjdk-1.8.0.262.b10-0.oe1.aarch64/jre
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "4.19.90-2110.8.0.0119.oe1.aarch64", arch: "aarch64", family: "unix"
```

执行以下命令， 进入Maven 配置文件目录

```
[root@ecs-oa maven]# cd /usr/local/maven/apache-maven-3.6.3/conf/
[root@ecs-oa conf]#
```

执行以下命令，　打开配置文件

```
<?xml version="1.0" encoding="UTF-8"?>

<!--
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or more contributor license agreements.  See the NOTICE file
distributed with this work for additional information
regarding copyright ownership.  The ASF licenses this file
to you under the Apache License, Version 2.0 (the
"License"); you may not use this file except in compliance
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    http://www.apache.org/licenses/LICENSE-2.0

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software distributed under the License is distributed on an
"AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY
KIND, either express or implied.  See the License for the
specific language governing permissions and limitations
under the License.
-->

<!--
 | This is the configuration file for Maven. It can be specified at two levels:
 |
 |  1. User Level. This settings.xml file provides configuration for a single user,
 |                 and is normally provided in ${user.home}/.m2/settings.xml.
 |
 |                 NOTE: This location can be overridden with the CLI option:
 |
 |                 -s /path/to/user/settings.xml
 |
 |  2. Global Level. This settings.xml file provides configuration for all Maven
 |                 users on a machine (assuming they're all using the same Maven
 |                 installation). It's normally provided in
 |                 ${maven.conf}/settings.xml.
 |
 |                 NOTE: This location can be overridden with the CLI option:
 |
 |                 -gs /path/to/global/settings.xml
 |
 | The sections in this sample file are intended to give you a running start at
 | getting the most out of your Maven installation. Where appropriate, the default
"settings.xml" [dos] 257L, 10468C
```

输入”:158”后按回车键，在<mirrors>和</mirrors>中间，插入以下代码。

```
  <mirrors>
    <!-- mirror
     | Specifies a repository mirror site to use instead of a given repository. The repository that
     | this mirror serves has an ID that matches the mirrorOf element of this mirror. IDs are used
     | for inheritance and direct lookup purposes, and must be unique across the set of mirrors.
     |
     <mirror>
      <id>mirrorId</id>
      <mirrorOf>repositoryId</mirrorOf>
      <name>Human Readable Name for this Mirror.</name>
      <url>http://my.repository.com/repo/path</url>
     </mirror>
     -->
<mirror>
<id>mirror</id>
<mirrorOf>*</mirrorOf>
<name>cmc-cd-mirror</name>
<url>https://mirrors.huaweicloud.com/repository/maven/ </url>
</mirror>

  </mirrors>
```

执行以下命令，安装 Git。

```
[root@ecs-oa conf]# dnf install -y git --nogpgcheck
Last metadata expiration check: 0:30:14 ago on Wed 01 May 2024 09:00:49 PM CST.
Dependencies resolved.
================================================================================
 Package                 Architecture        Version                 Repository      Size
================================================================================
Installing:
 git                     aarch64             2.27.0-3.oe1            update         6.7 M
Installing dependencies:
 perl-TermReadKey        aarch64             2.38-2.oe1                             24 k
 perl-Error              noarch              1:0.17028-1.oe1                        31 k
 perl-Git                noarch              2.27.0-3.oe1                           28 k

Transaction Summary
================================================================================
Install  4 Packages

Total size: 6.7 M
Installed size: 35 M
Downloading Packages:
[SKIPPED] perl-TermReadKey-2.38-2.oe1.aarch64.rpm: Already downloaded
[SKIPPED] git-2.27.0-3.oe1.aarch64.rpm: Already downloaded
[SKIPPED] perl-Error-0.17028-1.oe1.noarch.rpm: Already downloaded
[SKIPPED] perl-Git-2.27.0-3.oe1.noarch.rpm: Already downloaded
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : perl-Error-1:0.17028-1.oe1.noarch                      1/4
  Installing       : perl-TermReadKey-2.38-2.oe1.aarch64                    2/4
  Installing       : perl-Git-2.27.0-3.oe1.noarch                           3/4
  Installing       : git-2.27.0-3.oe1.aarch64                               4/4
  Running scriptlet: git-2.27.0-3.oe1.aarch64                               4/4
  Verifying        : perl-TermReadKey-2.38-2.oe1.aarch64                    1/4
  Verifying        : git-2.27.0-3.oe1.aarch64                               2/4
  Verifying        : perl-Error-1:0.17028-1.oe1.noarch                      3/4
  Verifying        : perl-Git-2.27.0-3.oe1.noarch                           4/4

Installed:
  git-2.27.0-3.oe1.aarch64          perl-TermReadKey-2.38-2.oe1.aarch64
  perl-Error-1:0.17028-1.oe1.noarch perl-Git-2.27.0-3.oe1.noarch
```

执行以下命令，验证Git 版本

```
Complete!
[root@ecs-oa conf]# git version
git version 2.27.0
```

执行以下命令， Clone 源码到本地。

```
[root@ecs-oa conf]# cd /home
[root@ecs-oa home]# git clone -b mysql-8 https://gitee.com/github-5407963/oasys_mysql.git
Cloning into 'oasys_mysql'...
remote: Enumerating objects: 1224, done.
remote: Total 1224 (delta 0), reused 0 (delta 0), pack-reused 1224
Receiving objects: 100% (1224/1224), 4.27 MiB | 2.05 MiB/s, done.
Resolving deltas: 100% (242/242), done.
```

执行以下命令， 进入 OA 系统数据库配置文件目录。

```
[root@ecs-oa home]# cd /home/oasys_mysql/src/main/resources/
[root@ecs-oa resources]#
```

执行以下命令，编辑应用配置文件。

```
[root@ecs-oa resources]#  vim application.properties
```

修改 url、username 和 password 为 ecs-mysql 的内网地址、mysql 的登录用户名和密码，保存文件，退出。




```
server.port=8088
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
spring.datasource.url=jdbc:mysql://192.168.0.210:3306/oasys?autoReconnect=true&useSSL=false&characterEncoding=utf-8&serverTimezone=Hongkong&rewriteBatchedStatements=true&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password=Jktzxcbnm1597538@
```

执行以下命令，进入OA 系统目录。

```
[root@ecs-oa resources]# cd /home/oasys_mysql/
[root@ecs-oa oasys_mysql]#
```

执行以下命令，重新 Maven 本地编译安装。

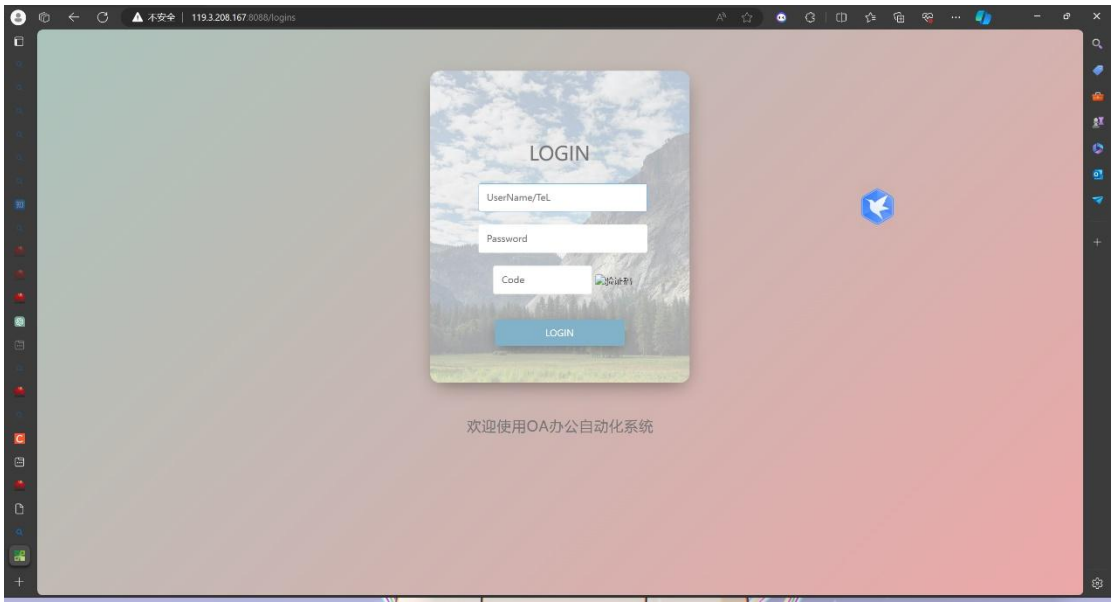
```
[root@ecs-oa oasys_mysql]# mvn install
[INFO] Scanning for projects...
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/boot/spring-boot-starter-parent/1.5.6.RELEASE/spring-boot-starter-parent-1.5.6.RELEASE.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/boot/spring-boot-starter-parent/1.5.6.RELEASE/spring-boot-starter-parent-1.5.6.RELEASE.pom (7.5 kB at 14 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/boot/spring-boot-dependencies/1.5.6.RELEASE/spring-boot-dependencies-1.5.6.RELEASE.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/boot/spring-boot-dependencies/1.5.6.RELEASE/spring-boot-dependencies-1.5.6.RELEASE.pom (93 kB at 379 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/com/fasterxml/jackson/jackson-bom/2.8.9/jackson-bom-2.8.9.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/com/fasterxml/jackson/jackson-bom/2.8.9/jackson-bom-2.8.9.pom (10 kB at 208 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/com/fasterxml/jackson/jackson-parent/2.8/jackson-parent-2.8.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/com/fasterxml/jackson/jackson-parent/2.8/jackson-parent-2.8.pom (8.0 kB at 67 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/com/fasterxml/oss-parent/27/oss-parent-27.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/com/fasterxml/oss-parent/27/oss-parent-27.pom (20 kB at 357 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/apache/logging/log4j/log4j-bom/2.7/log4j-bom-2.7.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/org/apache/logging/log4j/log4j-bom/2.7/log4j-bom-2.7.pom (5.4 kB at 37 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/apache/apache/9/apache-9.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/org/apache/apache/9/apache-9.pom (15 kB at 583 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/spring-framework-bom/4.3.10.RELEASE/spring-framework-bom-4.3.10.RELEASE.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/spring-framework-bom/4.3.10.RELEASE/spring-framework-bom-4.3.10.RELEASE.pom (5.1 kB at 41 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/data/spring-data-releasetrain/Ingalls-SR6/spring-data-releasetrain-Ingalls-SR6.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/data/spring-data-releasetrain/Ingalls-SR6/spring-data-releasetrain-Ingalls-SR6.pom (4.6 kB at 118 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/data/build/spring-data-build/1.9.6.RELEASE/spring-data-build-1.9.6.RELEASE.pom
Downloaded from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/data/build/spring-data-build/1.9.6.RELEASE/spring-data-build-1.9.6.RELEASE.pom (6.4 kB at 33 kB/s)
Downloading from mirror: https://mirrors.huaweicloud.com/repository/maven/org/springframework/integration/spring-integration-bom/4.3.11.RELEASE/spring-integration-bom-4.3.11.RELEASE.pom
```

执行以下命令，启动项目应用。

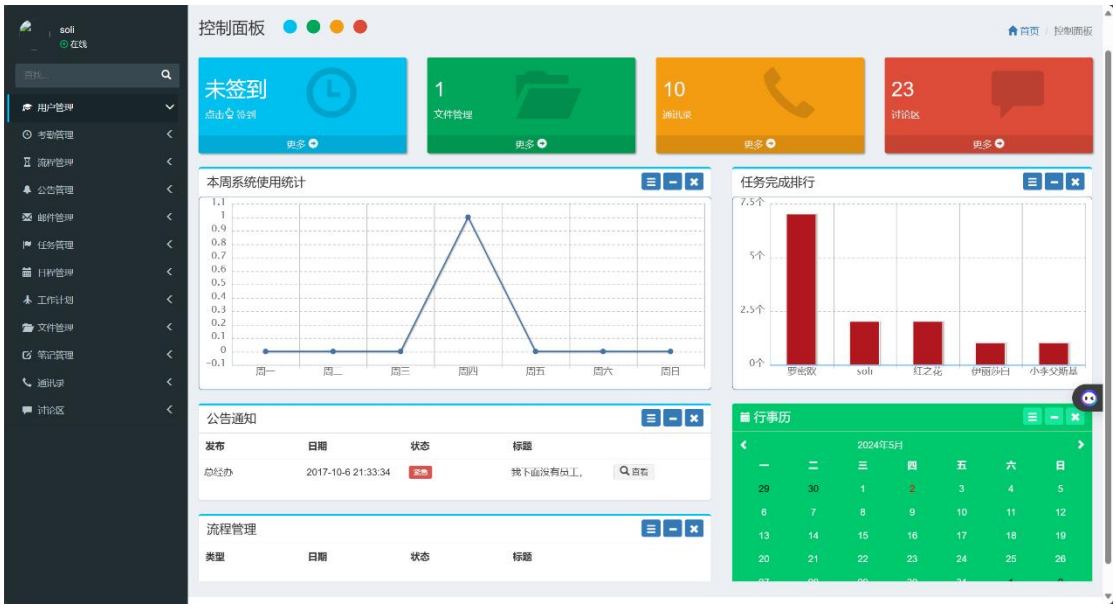
```
[root@ecs-oa oasys_mysql]# java -jar target/oasys.jar
http://www.network-science.de/ascii/
https://www.degraeve.com/img2txt.php
                                 /  /\         /  /\         /  /\          __             /  /\
                                /  ::\       /  ::\       /  ::\        |  |\           /  ::\
                               /  /:/\:\     /  /:/\:\     /__/:/\:\       |  |:|         /__/:/\:\
                              /  /::/  \:\   /  /::\ \:\   _\_ \:\ \:\      |  |:|        _\_ \:\ \:\
                             /__/:/ \__\:\ /__/:/\:\_\:\ /__/\ \:\ \:\     |__|:|__    /__/\ \:\ \:\
                             \  \:\ /  /:/ \__\/  \:\/:/ \  \:\ \:\_\/     /  /::::\ \  \:\ \:\_\/
                              \  \:\  /:/       \__\::/   \  \:\_\:\      /  /:/~~~~  \  \:\_\:\
                               \  \:\/:/        /  /:/     \  \:\/:/     /__/:/         \  \:\/:/
                                \  \::/        /__/:/       \  \::/      \__\/           \  \::/
                                 \__\/         \__\/         \__\/                       \__\/
                               Copyright (c) 2018 aaluoxiang  https://gitee.com/aaluoxiang/oa_system

2024-05-01 21:43:07.091  INFO 6419 --- [           main] cn.gson.oasys.OasysApplication           : Starting OasysApplication v0.0.1-SNAPSHOT on ecs-oa with PID 6419 (/home/oasys_mysql/target/oasys.jar started by root in /home/oasys_mysql)
2024-05-01 21:43:07.102  INFO 6419 --- [           main] cn.gson.oasys.OasysApplication           : No active profile set, falling back to default profiles: default
2024-05-01 21:43:07.218  INFO 6419 --- [           main] ationConfigEmbeddedWebApplicationContext : Refreshing org.springframework.boot.context.embedded.AnnotationConfigEmbeddedWebApplicationContext@6e5e91e4: startup date [Wed May 01 21:43:07 CST 2024]; root of context hierarchy
2024-05-01 21:43:10.861  INFO 6419 --- [           main] s.b.c.e.t.TomcatEmbeddedServletContainer : Tomcat initialized with port(s): 8088 (http)
2024-05-01 21:43:10.883  INFO 6419 --- [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2024-05-01 21:43:10.884  INFO 6419 --- [           main] org.apache.catalina.core.StandardEngine  : Starting Servlet Engine: Apache Tomcat/8.5.16
2024-05-01 21:43:11.013  INFO 6419 --- [ost-startStop-1] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
2024-05-01 21:43:11.014  INFO 6419 --- [ost-startStop-1] o.s.web.context.ContextLoader            : Root WebApplicationContext: initialization completed in 3799 ms
2024-05-01 21:43:11.236  INFO 6419 --- [ost-startStop-1] o.s.b.w.servlet.ServletRegistrationBean  : Mapping servlet: 'dispatcherServlet' to [/]
2024-05-01 21:43:11.241  INFO 6419 --- [ost-startStop-1] o.s.b.w.servlet.FilterRegistrationBean   : Mapping filter: 'characterEncodingFilter' to: [/*]
2024-05-01 21:43:11.241  INFO 6419 --- [ost-startStop-1] o.s.b.w.servlet.FilterRegistrationBean   : Mapping filter: 'hiddenHttpMethodFilter' to: [/*]
2024-05-01 21:43:11.241  INFO 6419 --- [ost-startStop-1] o.s.b.w.servlet.FilterRegistrationBean   : Mapping filter: 'httpPutFormContentFilter' to: [/*]
2024-05-01 21:43:11.241  INFO 6419 --- [ost-startStop-1] o.s.b.w.servlet.FilterRegistrationBean   : Mapping filter: 'requestContextFilter' to: [/*]
2024-05-01 21:43:12.068  INFO 6419 --- [           main] j.LocalContainerEntityManagerFactoryBean : Building JPA container EntityManagerFactory for persistence unit 'default'
2024-05-01 21:43:12.088  INFO 6419 --- [           main] o.hibernate.jpa.internal.util.LogHelper  : HHH000204: Processing PersistenceUnitInfo [
        name: default
        ...]
2024-05-01 21:43:12.343  INFO 6419 --- [           main] org.hibernate.Version                    : HHH000412: Hibernate Core {5.0.12.Final}
```

通过 ecs-oa 弹性公网 IP 地址加 8088 端口访问 OA 系统登录页

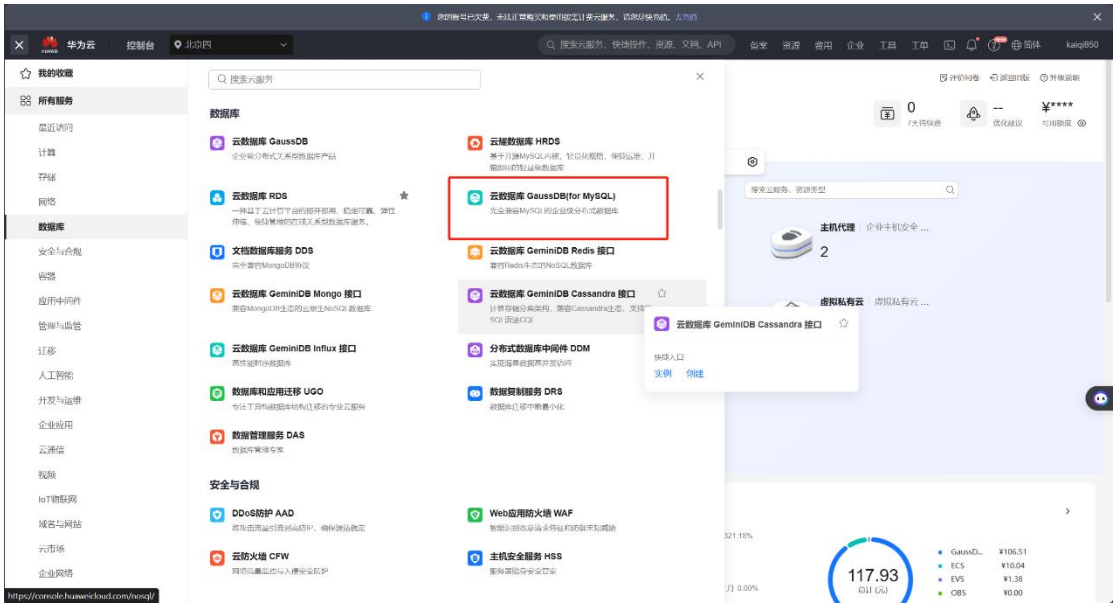


输入账号密码和验证码成功进入

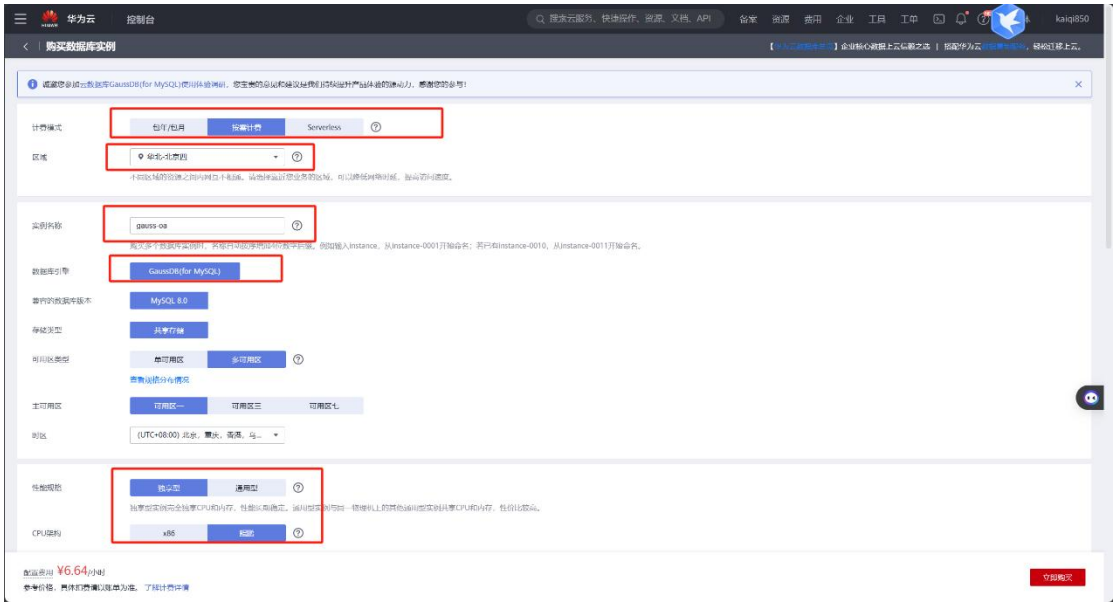


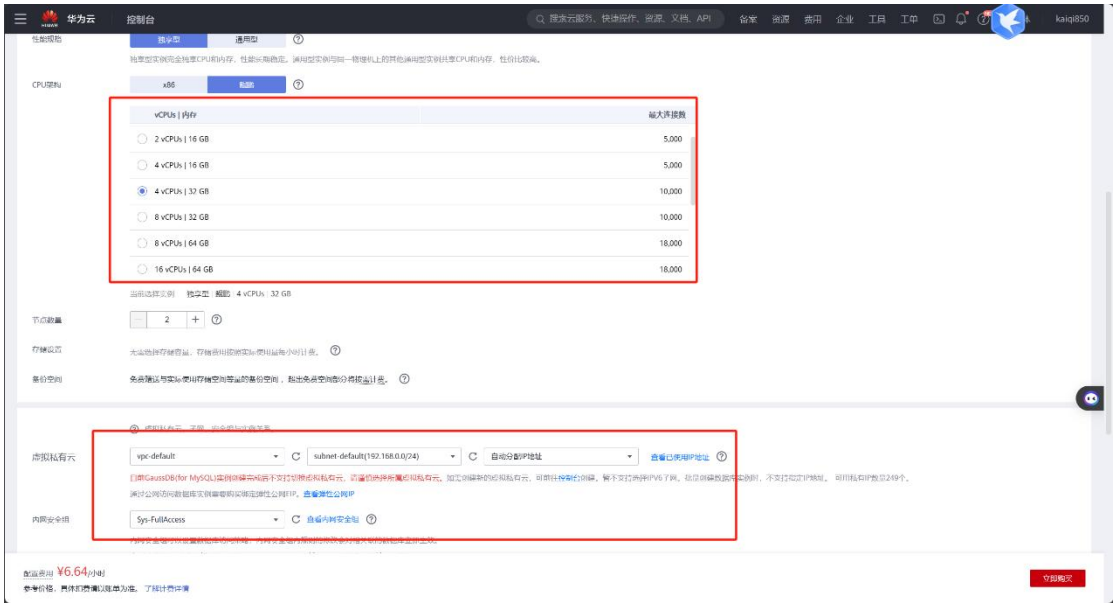
七．部署云数据库 GaussDB

在页面左上角，选择区域“北京四”，单击“服务列表”，选择“数据库 > 云数据库 GaussDB”，进入服务列表页面。



进入云数据库 GaussDB 基础配置页面，选择配置如下所示，点击“立即购买”。设置密码为：Jklzxcbnm1597538@



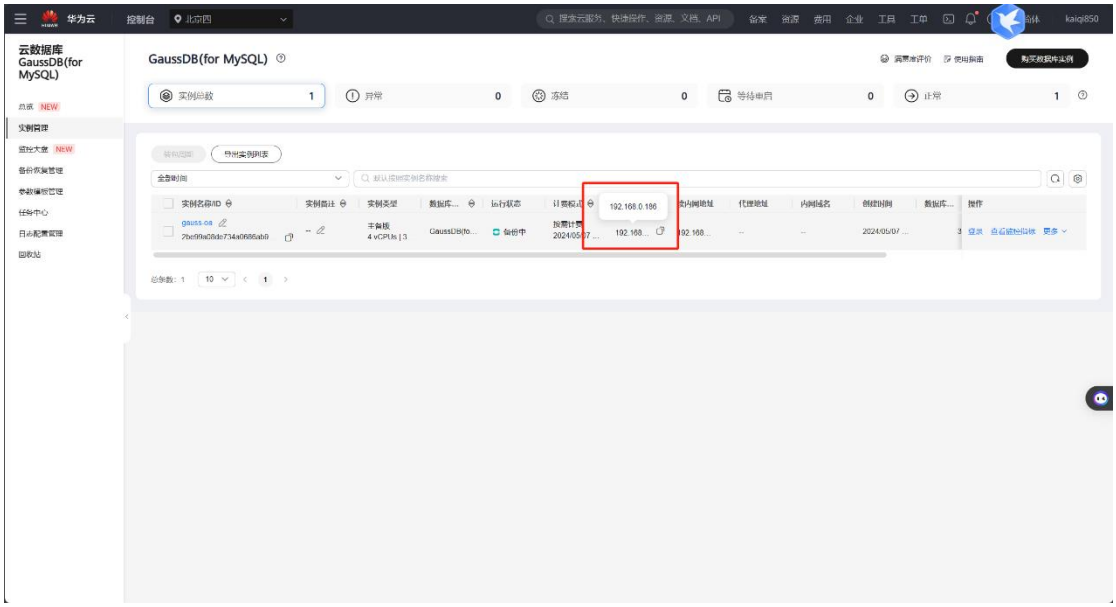


八．迁移数据

执行以下命令，导出数据库。

```
[root@ecs-e3b1 ~]# mysqldump -uroot -p oasys > oasys-dump.sql
Enter password:
[root@ecs-e3b1 ~]# ls
oasys-dump.sql
[root@ecs-e3b1 ~]#
```

通过 MySQL 客户端登录 gauss-oa ，其中 192.168.0.XXX 修改为 gauss-oa 相应的内网地址




```
[root@ecs-e3b1 ~]# mysqldump -uroot -p oasys > oasys-dump.sql
Enter password:
[root@ecs-e3b1 ~]# ls
oasys-dump.sql
[root@ecs-e3b1 ~]# mysqldump -uroot -p oasys > oasys-mysql.sql
Enter password:
[root@ecs-e3b1 ~]# ls
oasys-dump.sql  oasys-mysql.sql
[root@ecs-e3b1 ~]# mysql -h 192.168.0.186 -P 3306 -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 378
Server version: 8.0.22-30 Source distribution

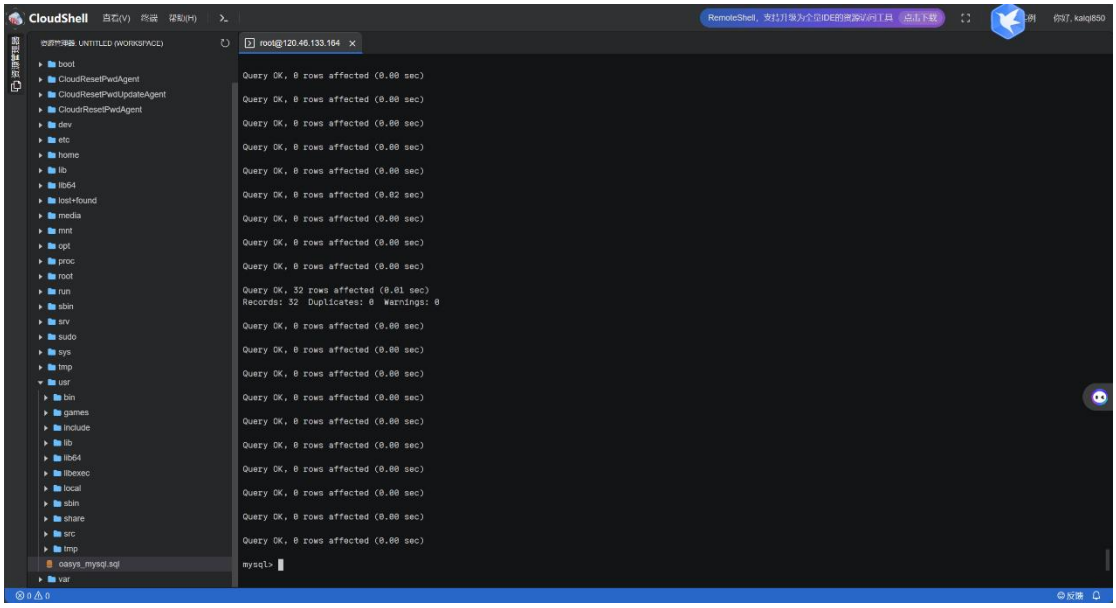
Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

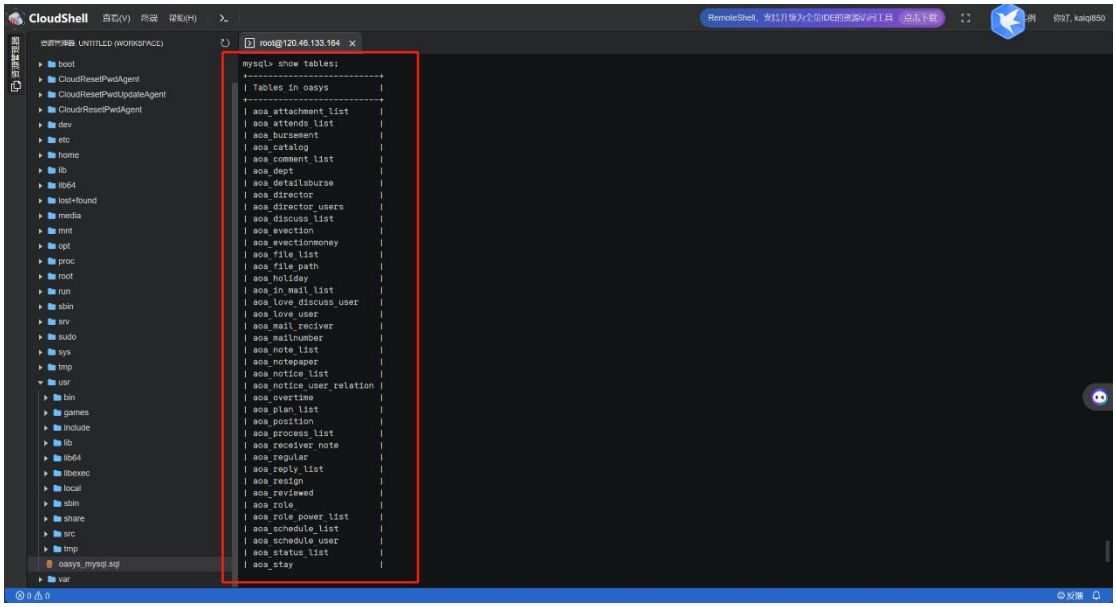
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

导入 oasys-mysql.sql 到 gauss-oa

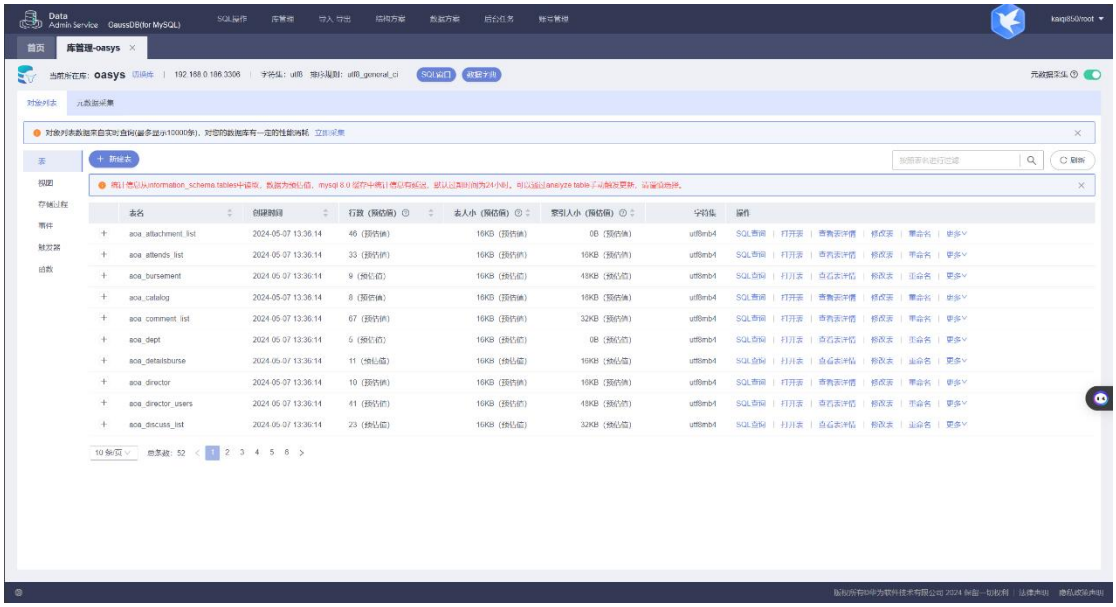
```
mysql> create database oasys;
Query OK, 1 row affected (0.01 sec)
```



查看数据库表并退出



也可以通过“登录 ”gauss-oa ，进入 DAS 控制台查看数据库表。



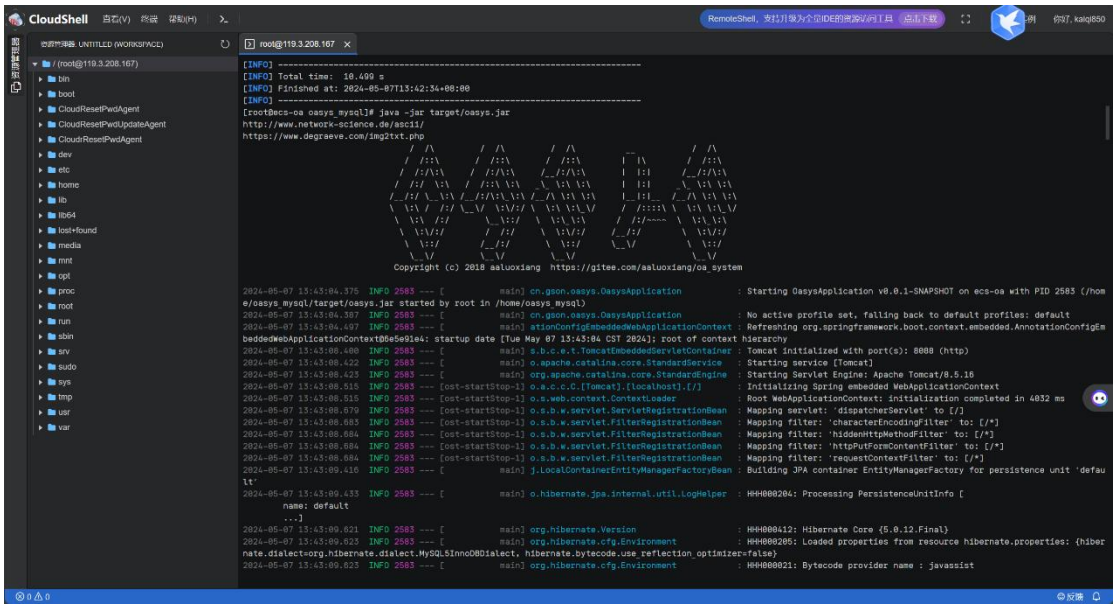
在使用 putty 登录 ecs-oa ，执行以下命令， 打开并修改数据库连接信息。 其中 IP 地址为 gauss-oa 的内网地址，密码为 gauss-oa 的 root 登录密码。

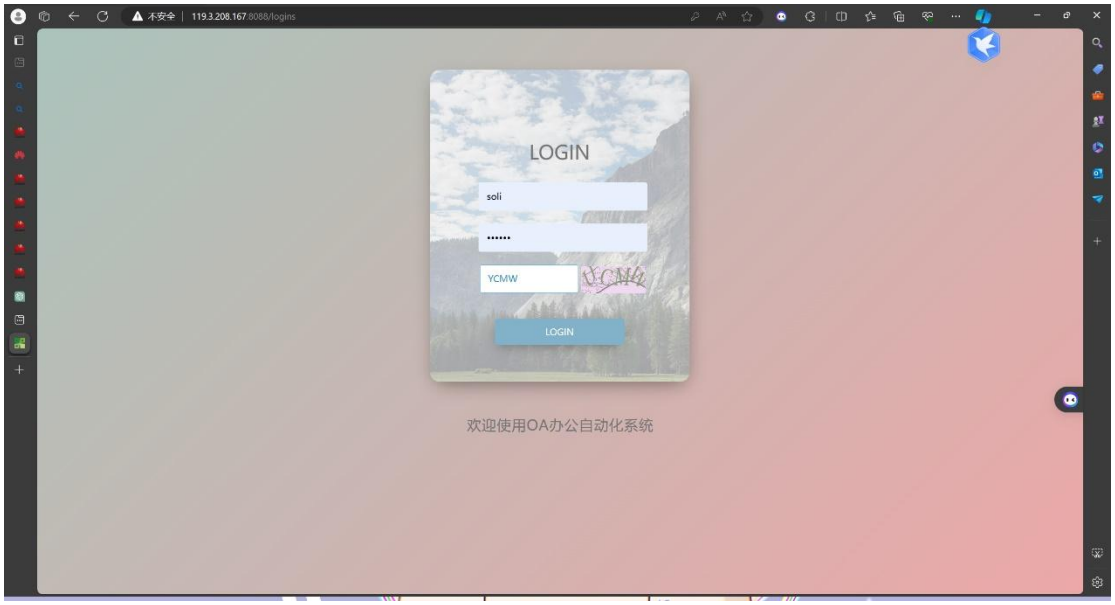
```
server.port=8888
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
spring.datasource.url=jdbc:mysql://192.168.0.186:3306/oasys?autoReconnect=true&useSSL=false&characterEncoding=utf-8&serverTimezone=Hongkong&rewriteBatchedStatements=true&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password=Jklzxcbnm1597538@
```

执行以下命令， 进入项目目录， 使用 Maven 本地安装。

```
[root@ecs-oa ~]#  cd /home/oasys_mysql/src/main/resources/
[root@ecs-oa resources]# vim application.properties
[root@ecs-oa resources]# cd /home/oasys_mysql/
[root@ecs-oa oasys_mysql]# mvn install
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------------< cn.gson:oasys >----------------------------
[INFO] Building oasys 0.0.1-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ oasys ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO] Copying 777 resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ oasys ---
```

执行以下命令，启动应用。通过 ecs-oa 的弹性公网 IP 地址加 8088 端口访问 OA 系统登录页， 使用用户名“soli”和密码“123456”登录 OA 系统主界面。

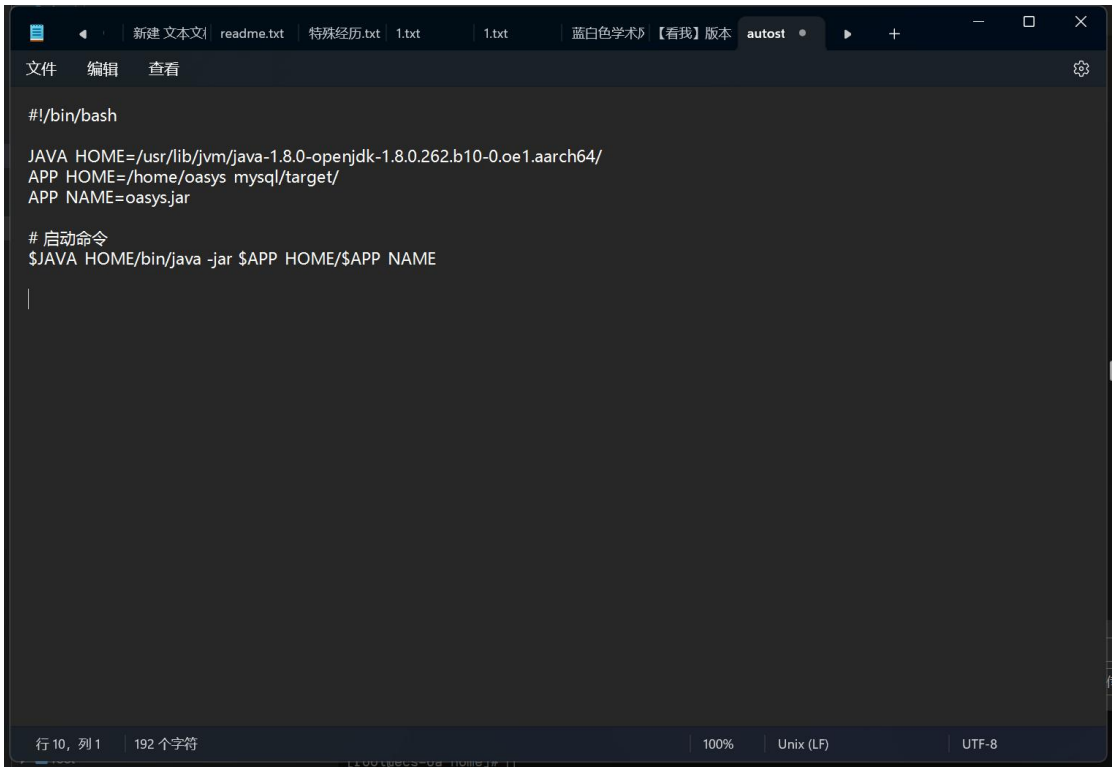





oasys
控制面板
未签到
1
文件管理
10
通讯录
23
讨论区
本周系统使用统计
任务完成排行
公告通知
流程管理
行事历

九．自启动设置

修改脚本


新建 文本文
readme.txt
特殊经历.txt
1.txt
1.txt
蓝白色学术
【看我】版本
autost
文件 编辑 查看
#!/bin/bash
JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-1.8.0.262.b10-0.oe1.aarch64/
APP_HOME=/home/oasys_mysql/target/
APP_NAME=oasys.jar
启动命令
$JAVA_HOME/bin/java -jar $APP_HOME/$APP_NAME
行 10，列 1
192 个字符
100%
Unix (LF)
UTF-8

执行以下命令，为脚本添加可执行权限。

```
[root@ecs-oa home]# chmod +x autostart.sh
[root@ecs-oa home]#
```

修改脚本

```
[Unit]
Description=My Spring Boot Service
After=syslog.target

[Service]
User=root
WorkingDirectory=/home/oasys mysql/target/
ExecStart=/usr/lib/jvm/java-1.8.0-openjdk-1.8.0.262.b10-0.oe1.aarch64/bin/java -jar oasys.jar
SuccessExitStatus=143

[Install]
WantedBy=multi-user.target
```

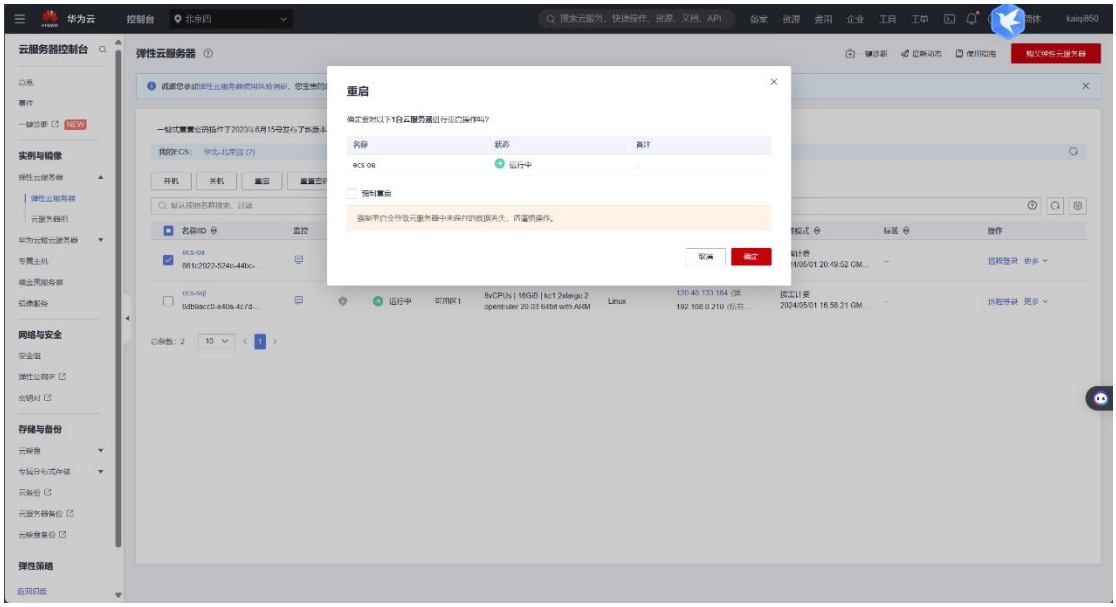
执行以下命令，为脚本添加可执行权限。

```
[root@ecs-oa system]# chmod +x oa-service.service
[root@ecs-oa system]#
```

执行以下命令，添加开机启动服务。

```
[root@ecs-oa system]# systemctl enable oa-service.service
Created symlink /etc/systemd/system/multi-user.target.wants/oa-service.service → /etc/systemd/system/oa-service.service.
[root@ecs-oa system]#
```

重启验证

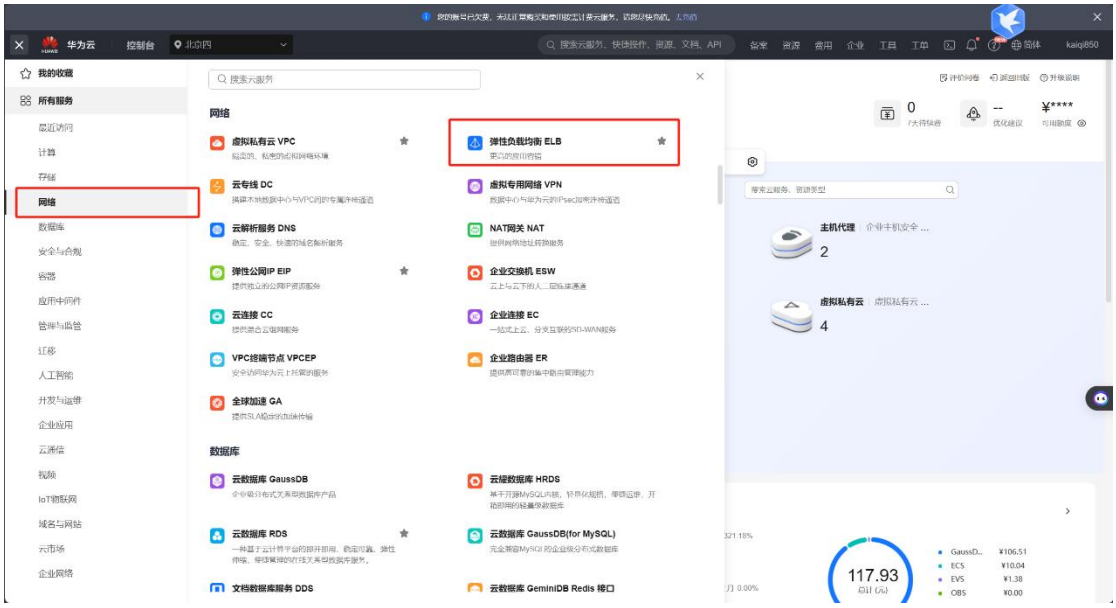


等待 1-2 分钟，通过 ecs-oa 弹性公网 IP 地址加 8088 端口访问 OA 系统登录页，如 http://124.70.8.248 :8088/ 。使用用户名“soli”和密码“123456”登录进入 OA 系统主界 面。

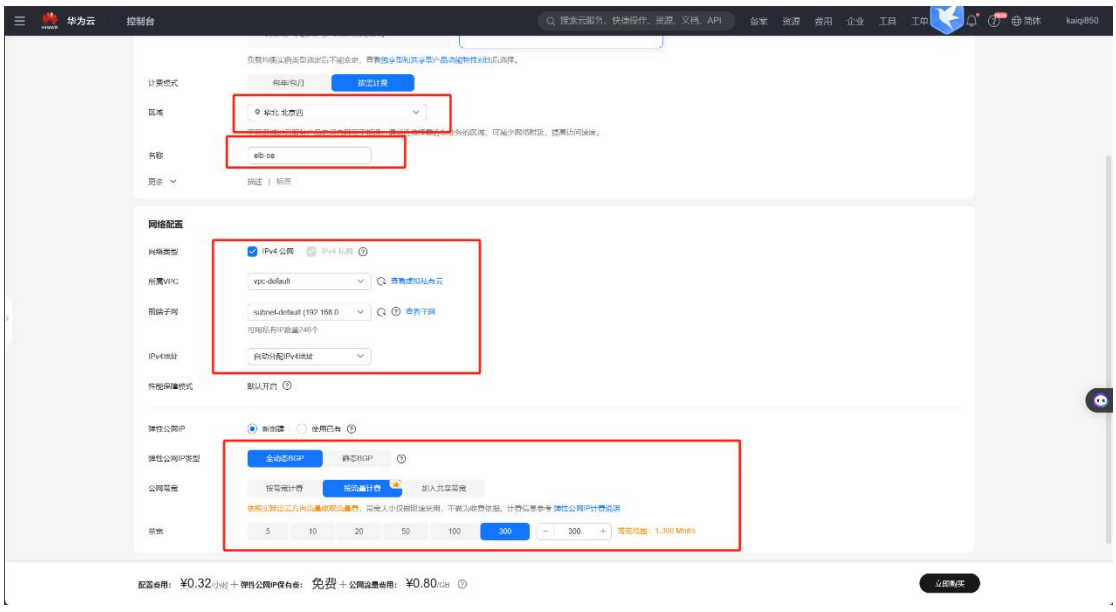


十．配置弹性负载均衡

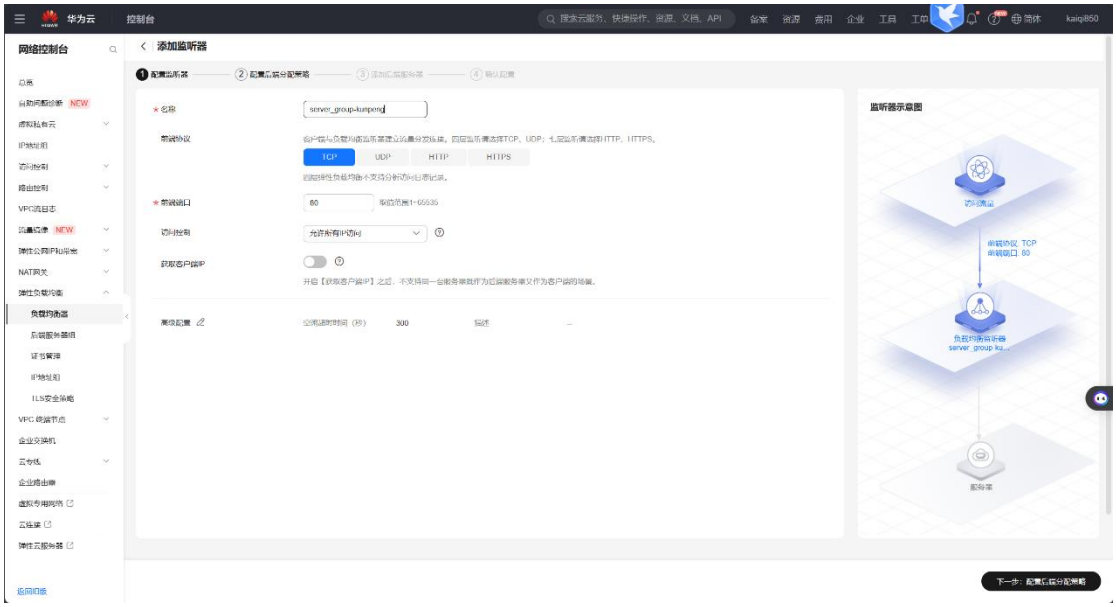
进入华为云控制台操作页面。 选择区域“北京四” ，单击“服务列表 ”，选择“网络 弹 性负载均衡 ELB” ，进入弹性负载均衡列表页面。



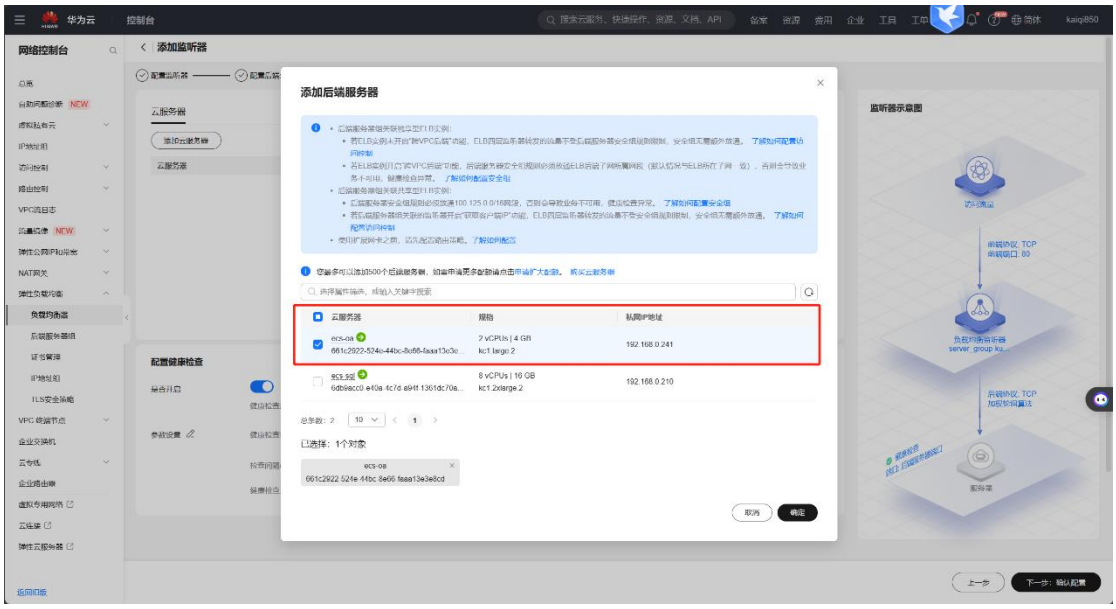
选择与 ecs-oa 相同的“所属 VPC”和“ 网络” ，选择“按流量计费”，输入名称“elb- oa” ，点击右下角“立即购买 ”。



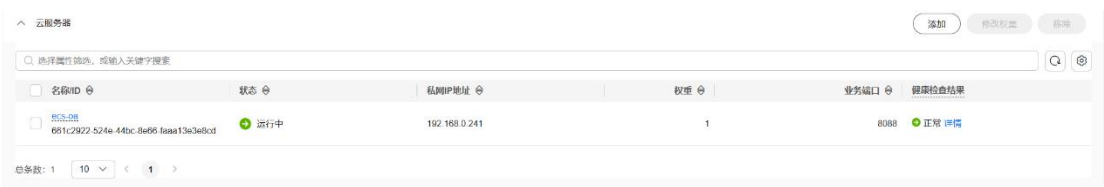
在负载均衡器列表中，点击“点我开始配置 ”。添加监听器



选择名为“ecs-oa”的云服务器， 点击“下一步”。



健康检查结果为正常



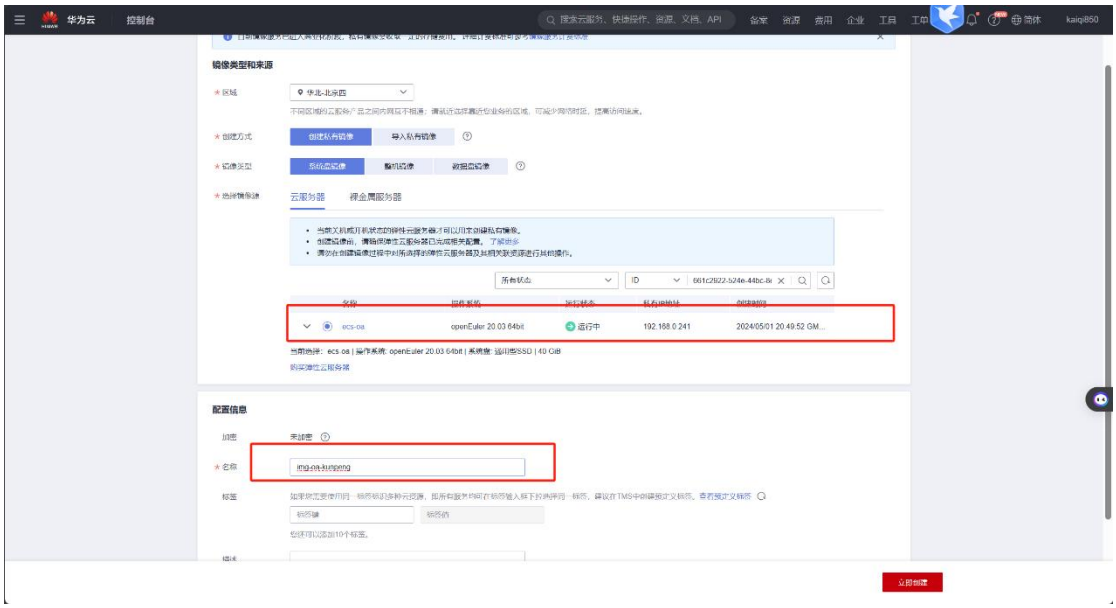
返回负载均衡器列表，记录“elb-oa”的弹性公网 IP ，使用该 IP 地址直接访问 OA 系统登录 页，如 http://119.3.211.28/ 。使用用户名“soli”和密码“123456”登录进入 OA 系统主 界面。



十一.创建鲲鹏云服务器镜像

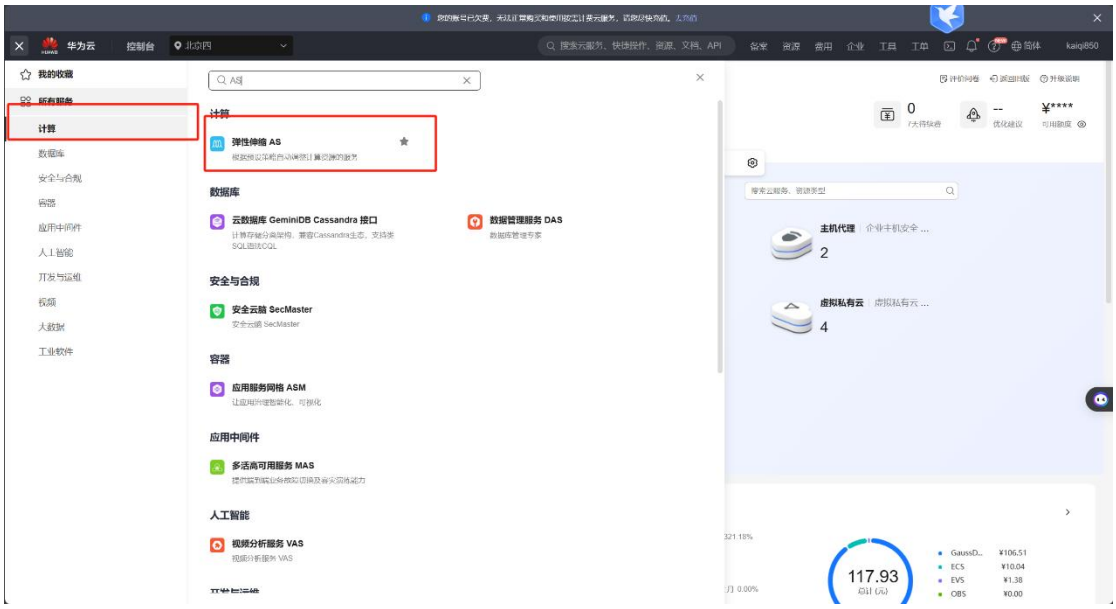
返回弹性云服务器列表， 点击“更多 -> 镜像/磁盘 -> 创建镜像 ”。

选择名为“ecs-oa”的云服务器。

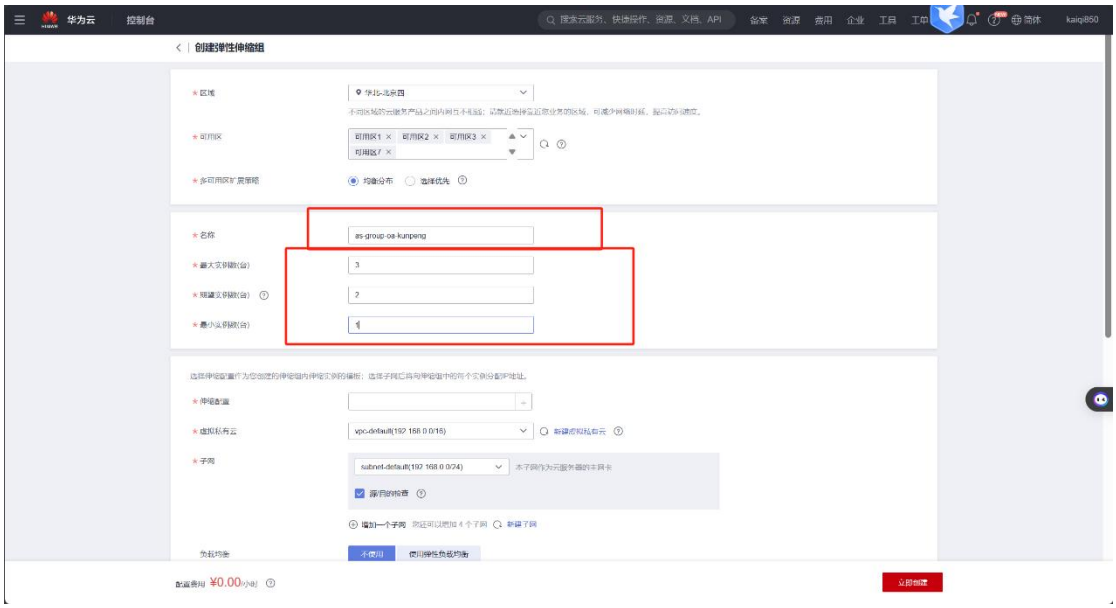


十二.购买弹性伸缩

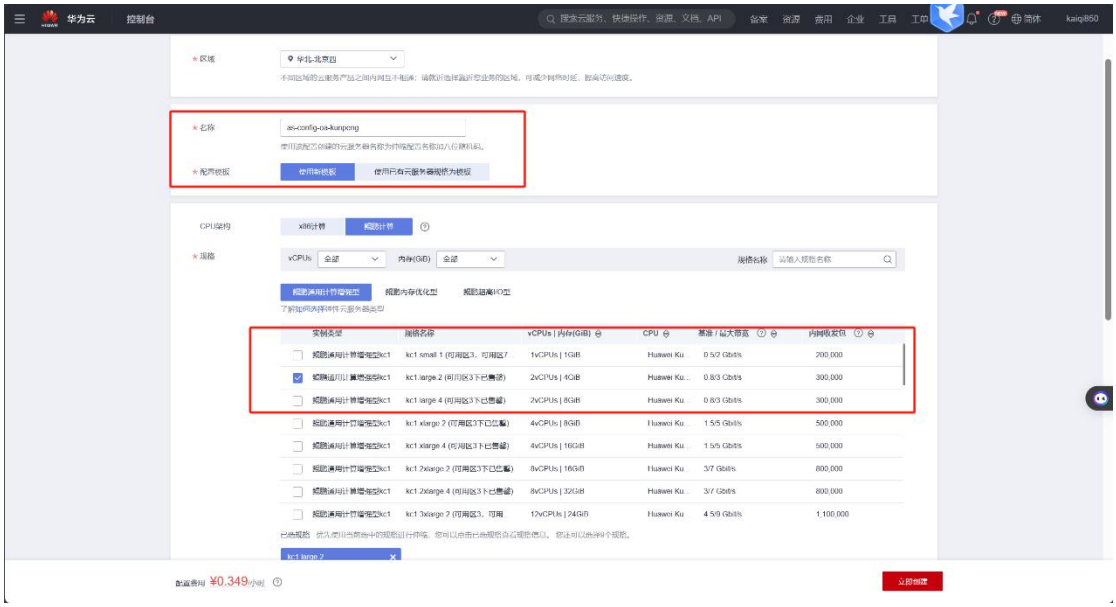
进入华为云控制台操作页面。 选择区域“北京四” ，单击“服务列表 ”，选择“弹性伸缩

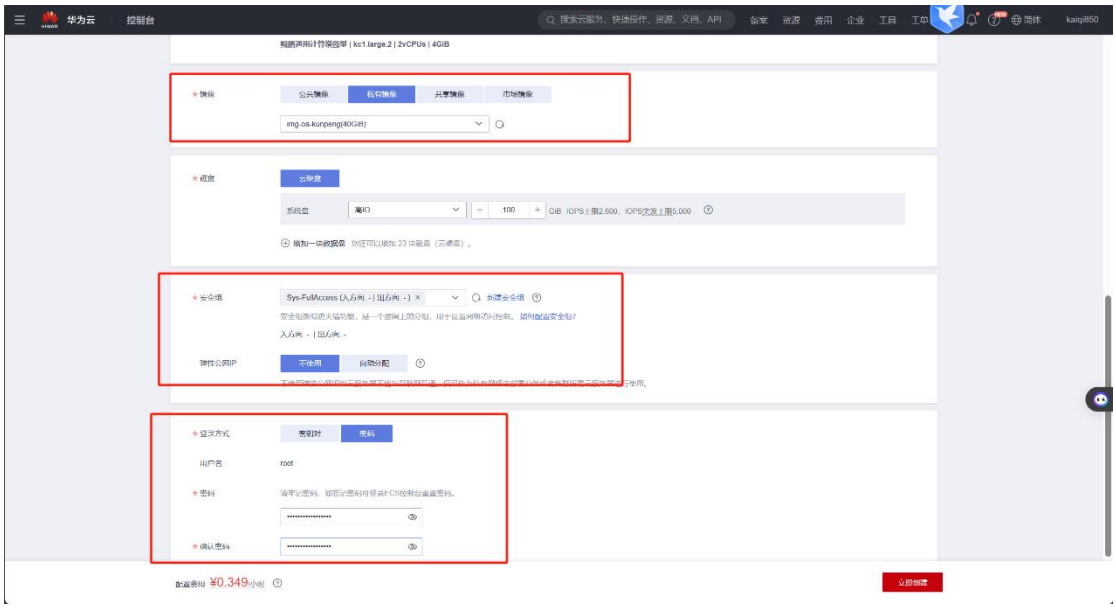


AS” ，进入弹性伸缩列表页面。

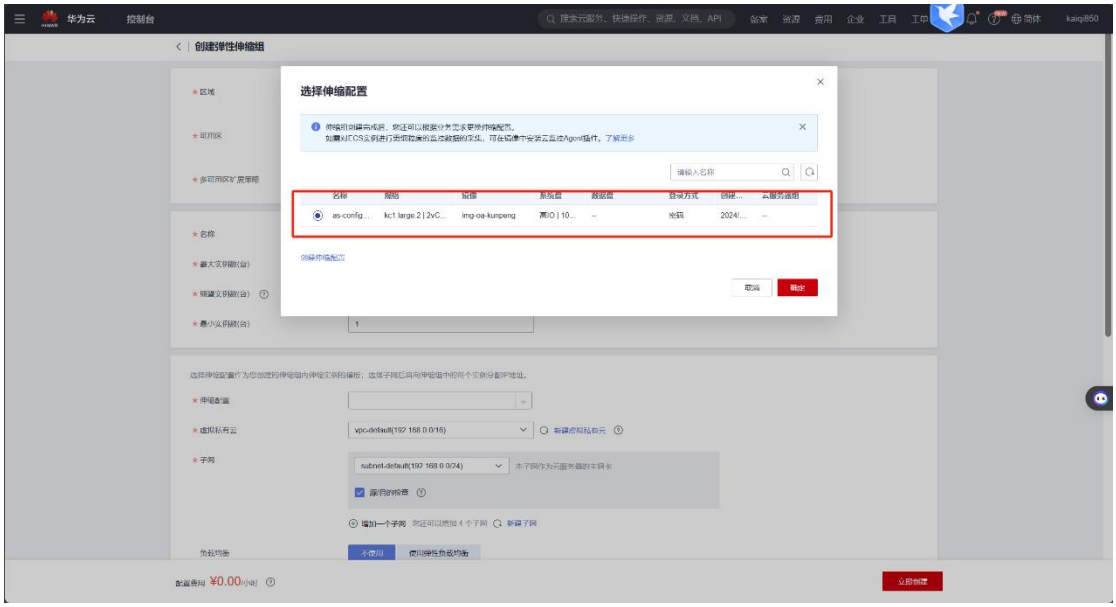


创建配置


华为云
控制台
kaiqi850
¥0.349/小时
立即创建


华为云
控制台
kaiqi850
root
¥0.349/小时
立即创建

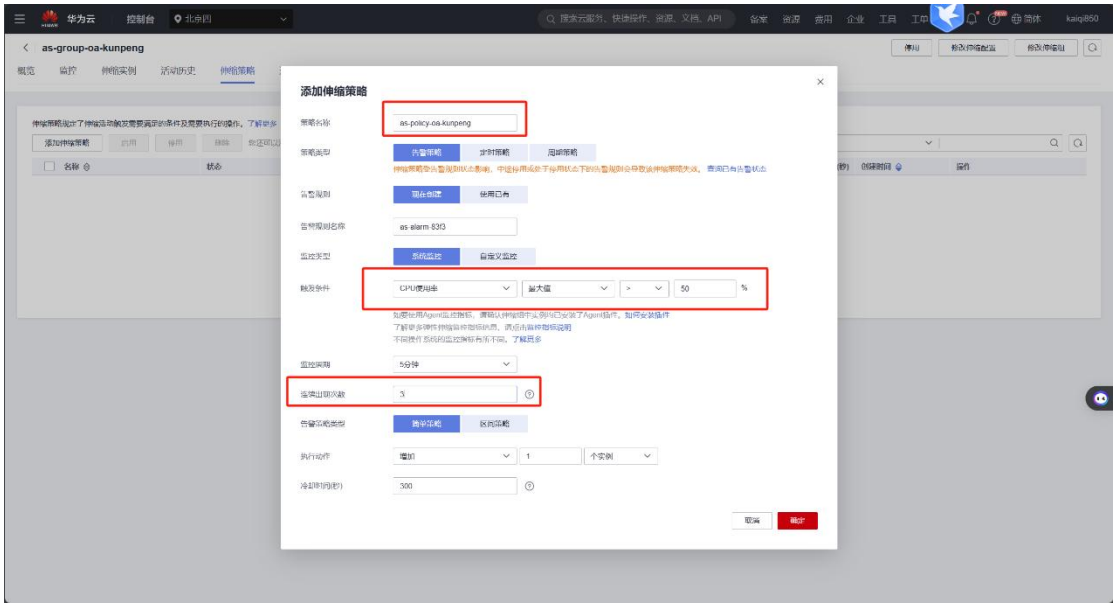
返回到选择伸缩配置页面，点击刷新按钮，选择“as-config-oa-kunpeng”的配置，点击“确定”。



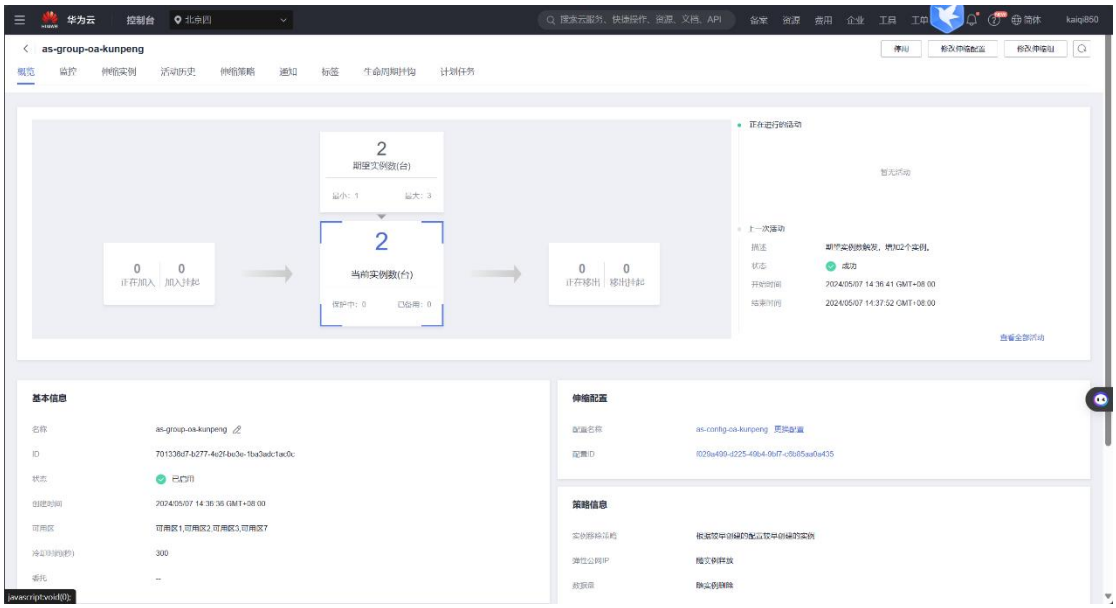
确认使用和 ecs-oa 同样的虚拟私有云和子网，负载均衡选择“使用弹性负载均衡”，负载均衡器选择“elb-oa”，后端服务器组选择“server_group-kunpeng”，后端端口填写“8088”，点击“立即创建”。



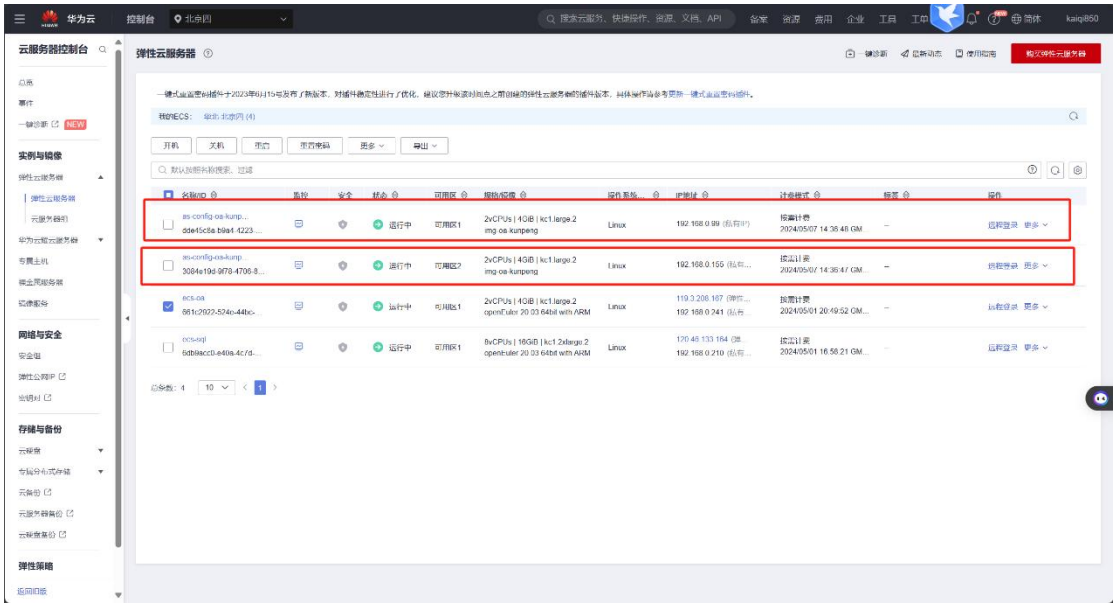
配置伸缩策略



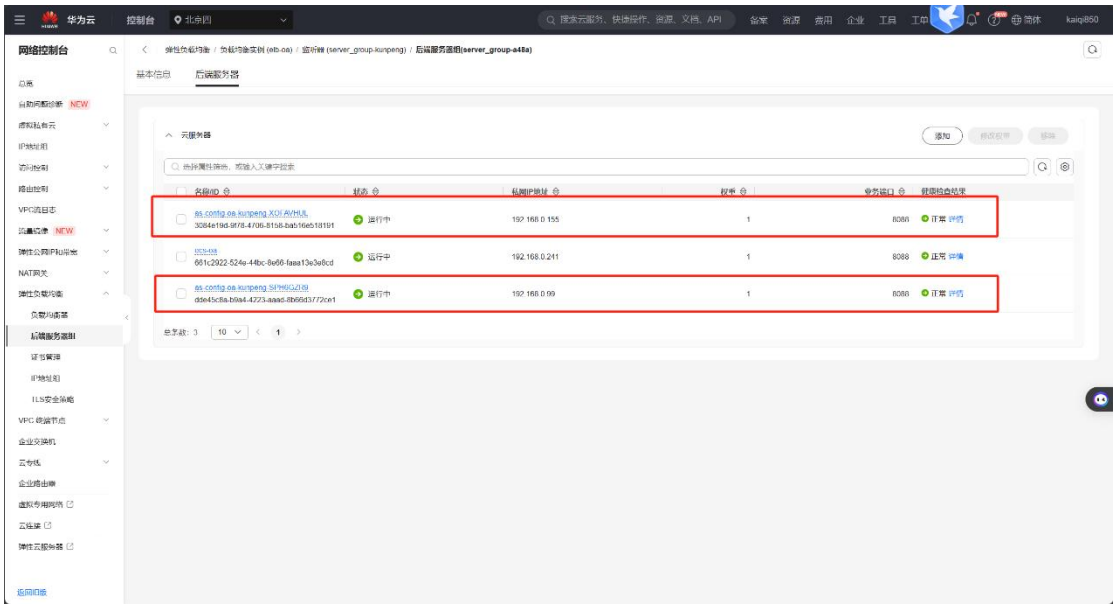
点击“概览”标签，显示当前实例数为 2 台， 即“期望实例数 ”。



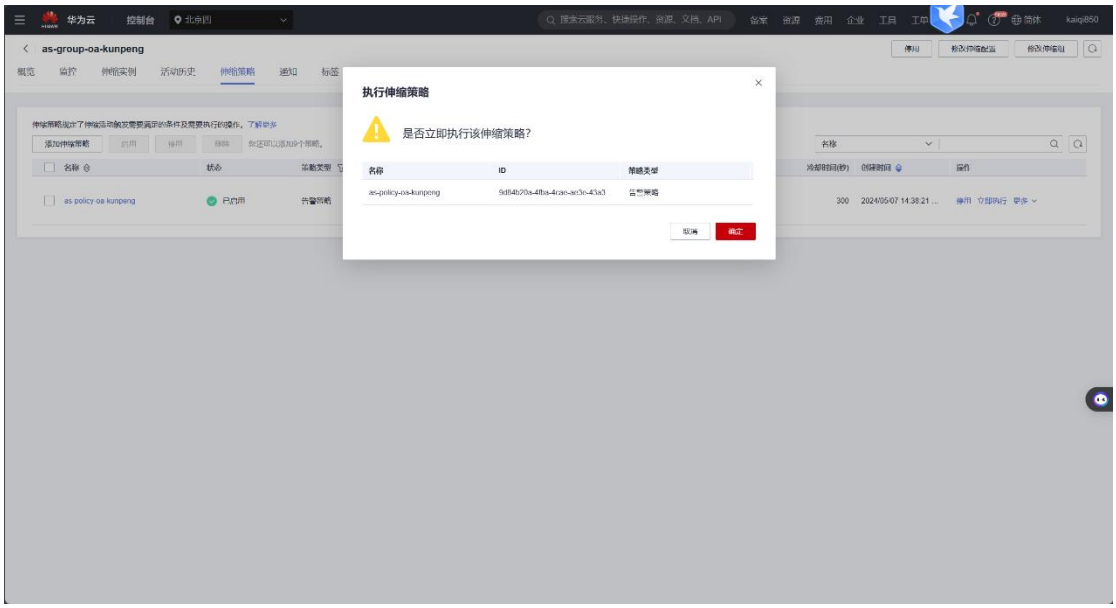
返回弹性云服务器列表， 点击刷新按钮，显示通过 AS 创建的鲲鹏实例。

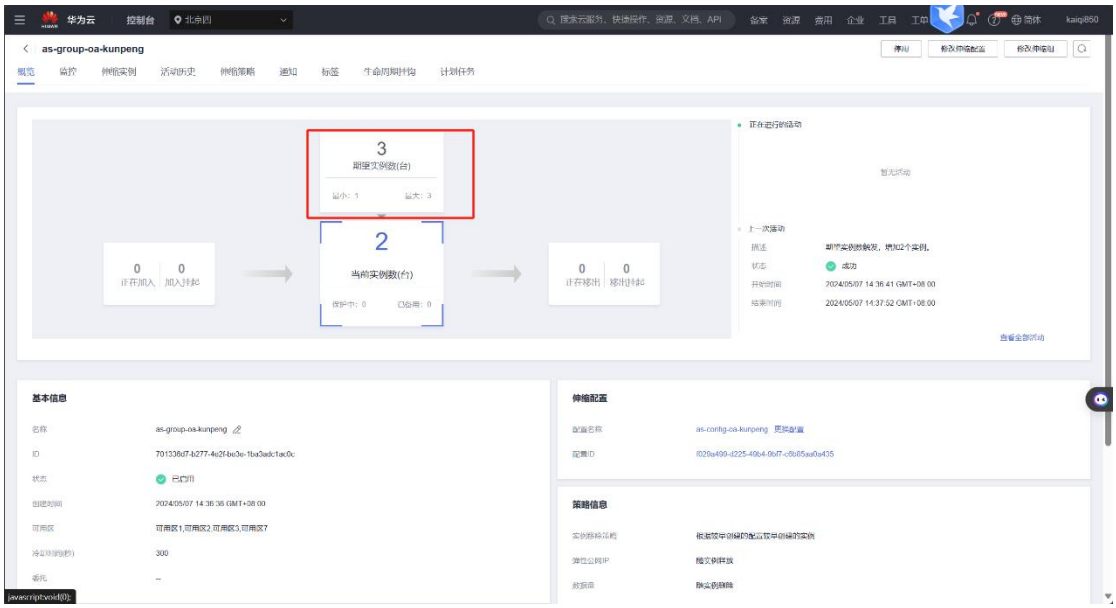


进入弹性负载均衡器“elb-oa”的后端服务器组”标签，显示这两台鲲鹏实例已加入。



返回伸缩实例 as-group-oa-kunpeng 的“伸缩策略 ”页面，点击“立即执行 ”，在对话框中点击“是 ”。在“概览”页面、云服务器列表和弹性负载均衡后端服务器组中都可显示新 创建的鲲鹏实例。





返回负载均衡器列表，记录“elb-oa”的弹性公网 IP，使用该 IP 地址直接访问 OA 系统登录页 。使用用户名“soli”和密码“123456”登录进入 OA 系统主界面。



十三.资源释放

删除 AS 弹性伸缩

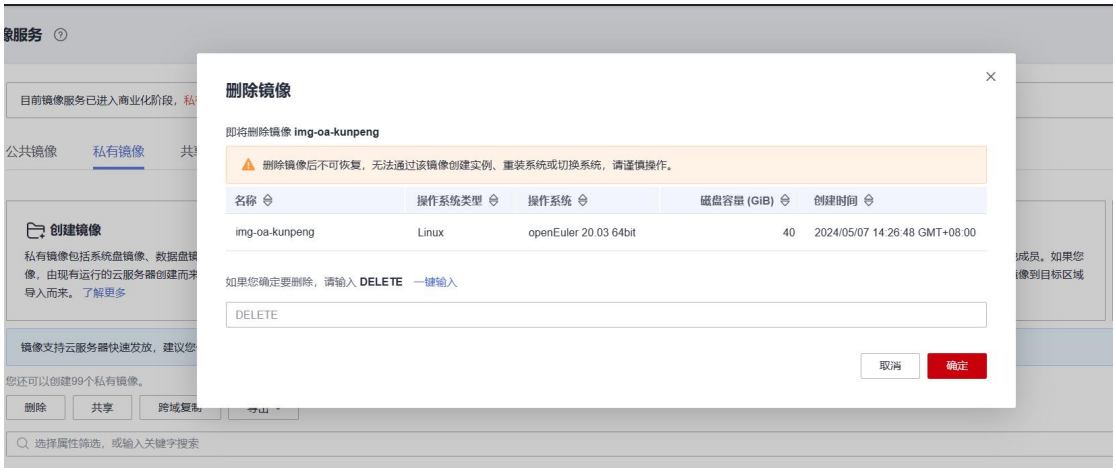


删除 ELB 弹性负载均衡




弹性负载均衡
新特性帮助视频
为提供更为稳定可靠的负载均衡服务
续费
转包年/包月
选择属性筛选，或输入关键字搜索
名称/ID
elb-oa
3eec2778-2dc8-4197-8b45-76
总条数：1
10
1
确认删除以下弹性负载均衡器吗？
确定删除后，负载均衡实例及其下配置（监听器、转发策略、后端服务器组等）无法恢复，后端服务器自动取消与后端服务器组的关联，请谨慎操作。
名称
状态
服务地址
可释放的弹性公网IP
删除监听器
删除后端服务器组
elb-oa
运行中
IPv4私有地址：192.168.0.228
IPv4公网地址：119.3.178.164
1
1
1
释放该负载均衡绑定的弹性公网IP，如不释放可能会被其他资源绑定继续计费。
如您确认要删除负载均衡实例及其下资源，请输入DELETE进行删除。
DELETE
否
是

删除 IMS 镜像


像服务
目前镜像服务已进入商业化阶段，
公共镜像
私有镜像
创建镜像
私有镜像包括系统盘镜像、数据盘镜像，由现有运行的云服务器创建而来或导入而来。了解更多
镜像支持云服务器快速发放，建议您
您还可以创建99个私有镜像。
删除
共享
跨域复制
选择属性筛选，或输入关键字搜索
删除镜像
即将删除镜像 img-oa-kunpeng
删除镜像后不可恢复，无法通过该镜像创建实例、重装系统或切换系统，请谨慎操作。
名称
操作系统类型
操作系统
磁盘容量 (GiB)
创建时间
img-oa-kunpeng
Linux
openEuler 20.03 64bit
40
2024/05/07 14:26:48 GMT+08:00
如果您确定要删除，请输入 DELETE 一键输入
DELETE
取消
确定

五、实验数据（现象）处理分析

问题与解决：

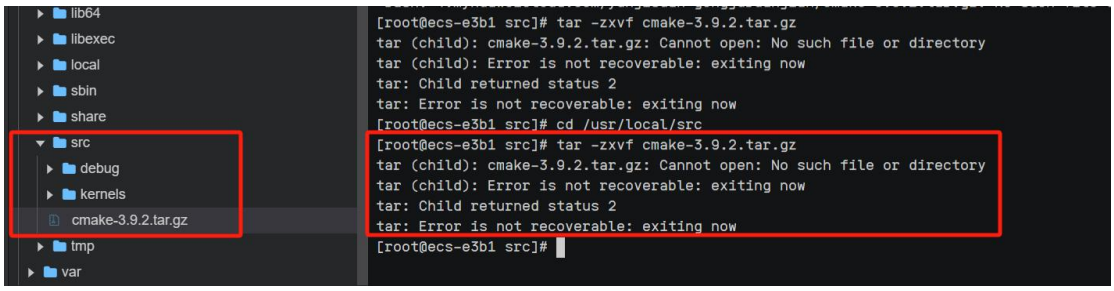
Q1：在安装 Cmake 时，使用指导书的指令，发现给的网站下载地址已经过期

```
[root@ecs-e3b1 ~]# cd /usr/local/src
[root@ecs-e3b1 src]# wget https://zhuanyejianshe.obs.cn-north-
--2024-05-01 17:11:47--  https://zhuanyejianshe.obs.cn-north-/
Resolving zhuanyejianshe.obs.cn-north- (zhuanyejianshe.obs.cn-north-)... failed: Name or service not known.
wget: unable to resolve host address 'zhuanyejianshe.obs.cn-north-'
[root@ecs-e3b1 src]# 4.myhuaweicloud.com/yunjisuan-gongjuruanjian/cmake-3.9.2.tar.gz
-bash: 4.myhuaweicloud.com/yunjisuan-gongjuruanjian/cmake-3.9.2.tar.gz: No such file or directory
```

S1：使用压缩包中的 Cmake 压缩包，上传到云端进行手动下载。



Q2:已经成功上传压缩包，解压时却提示找不到文件



S2：发现进入的 src 不是当前所在位置的 src，更换所在位置，成功下载。

Q3:使用 yum install mysql80-community-release-el8-1.noarch.rpm

命令安装的 SQL 是社区版，不会在本地留下任何文件，因此指导书提到的安装数据库到/home 的位置，产生了问题

S3：发现安装到的位置不一定要在 home 位置，只要在主机处就行。

Q4：

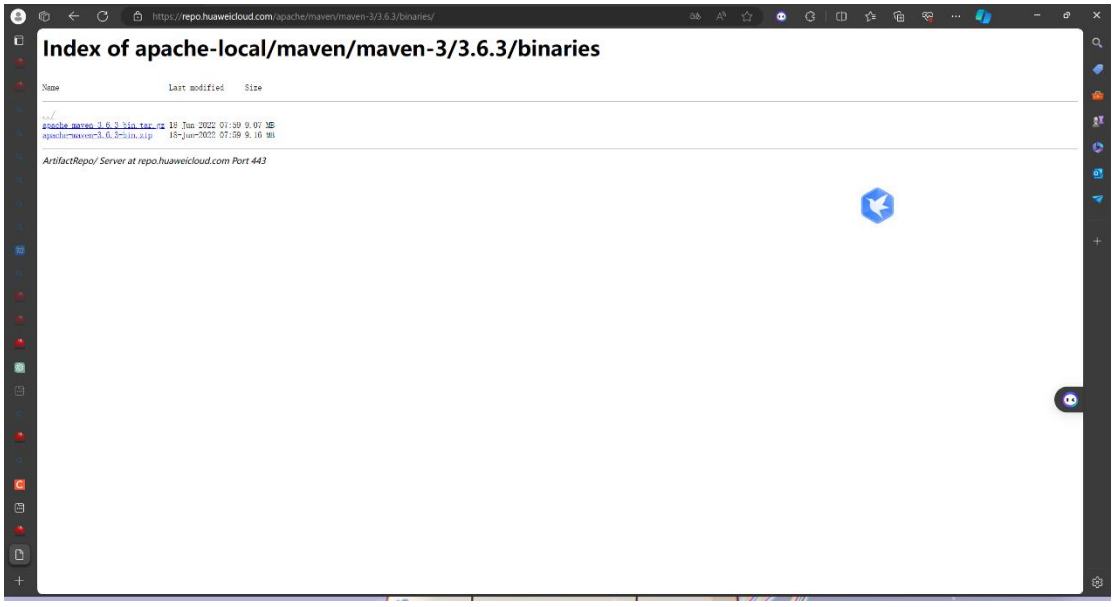
执行以下命令，获取 Maven 二进制包。↵

```
[root@ecs-oa maven]# wget http://mirrors.huaweicloud.com/apache/maven/maven-3/3.6.3/binaries/apache-maven-3.6.3-bin.tar.gz↵
```

给出的网站过期，且资源包中没有对应资源

S4：

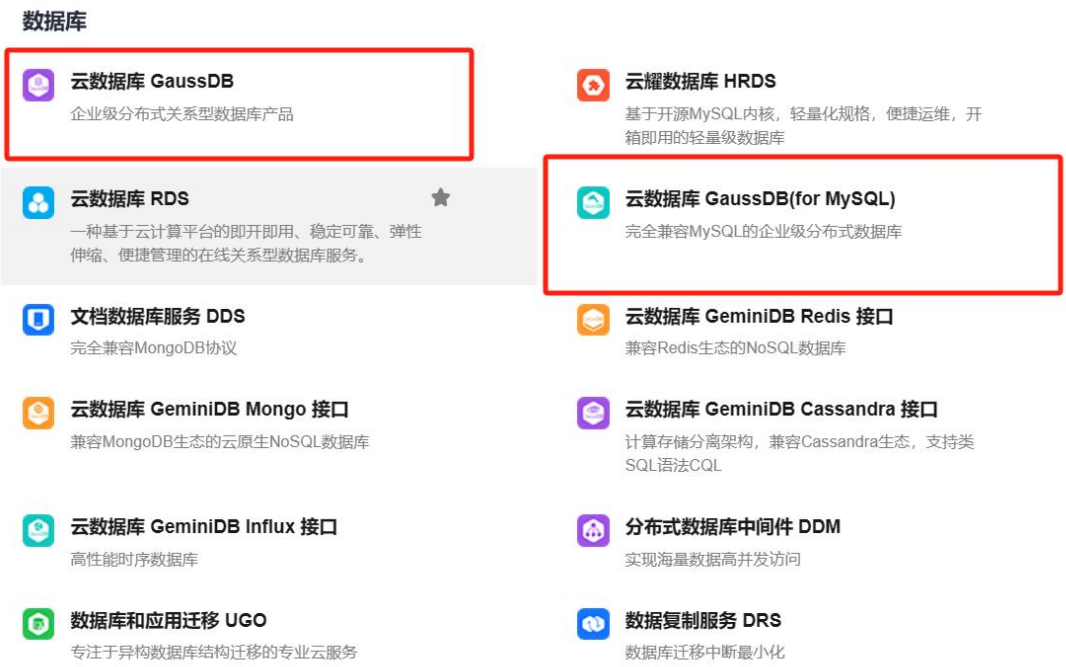
手动找网站，寻找对应资源



Q5：连接 Guass 数据库时显示无法连接

```
[root@ecs-e3b1 ~]# mysql -h 192.168.0.230 -P 3306 -u root -p
Enter password:
ERROR 2003 (HY000): Can't connect to MySQL server on '192.168.0.230:3306' (113)
```

S5:发现购买的数据库不适配弹性云服务器，更换数据库后成功连接



六、实验结论

通过本次实验，不仅成功完成了实验目标，且收获了大量知识和经验。

实验成果：

- 成功在华为云鲲鹏平台上部署了企业自动化办公（OA）系统。
- 将本地自建的 MySQL 数据库成功迁移到云数据库 GaussDB。
- 在鲲鹏平台上实现了应用的高可用部署，包括配置了弹性负载均衡和弹性伸缩服务。
- 最终成功搭建了一个完整的 OA 系统，并且确保了其在云平台上的稳定运行。

从中学习到的知识：

- 掌握了华为云鲲鹏云服务的操作方法，包括弹性云服务器 ECS 的部署和配置，云数据库 GaussDB 的部署和配置，弹性负载均衡 ELB 和弹性伸缩 AS 服务的配置等。
- 了解了如何将本地自建的 MySQL 数据库迁移到云数据库 GaussDB，并掌握了相关的原理、流程和操作方法。

- 学习了应用在云平台上的高可用部署方法，包括配置弹性负载均衡和弹性伸缩服务，以确保系统的稳定性和可靠性。

克服的问题：

- 遇到 CMake 下载地址过期的问题,但通过手动下载 CMake 压缩包并上传到云端解决了这一问题。
- 在安装 MySQL 数据库时，初始安装的版本为社区版，导致在本地留下的文件不符合预期，但通过重新安装并选择正确的版本解决了这一问题。
- 在连接 Gauss 数据库时出现无法连接的问题，经过检查发现是数据库版本不兼容导致的，最终成功解决了这一问题，并完成了数据迁移。

综上所述，通过这次实验，本人不仅完成了实验报告中所述的各项任务，还学到了如何在云平台上部署和管理应用，以及如何解决在实验过程中遇到的各种问题。这些知识和经验对于今后在云计算领域的工作和学习将会大有裨益。

七、教师批阅意见：

成绩评定：

教师签字：

年　　月　　日

八、备注：